# From Brains to Models: What AI Reveals About Being Human

Arvid Lundervold BSc, MD, PhD
Department of Biomedicine, University of Bergen
Mohn Medical Imaging and Visualization Center

Draft — compiled 2026-08-20 18:13:55+02:00

**Abstract**

The arrival of foundation models—large language models (LLMs) and large multimodal models (LMMs)—has turned a philosophical question into an empirical one: what, exactly, is distinctive about human cognition? This perspective examines what modern AI reveals about being human by holding two systems up against each other: the brain, increasingly understood as a prediction machine operating far from equilibrium, and deep artificial networks trained by large-scale optimization. We organize the comparison along the classical axes of *thinking humanly*, *thinking rationally*, *acting humanly*, and *acting rationally*, and along three levels of description: architecture (neurons and synapses versus attention and weights), dynamics (multistable dynamical systems versus gradient flows on loss landscapes), and function (prediction, agency, and value). We argue that the most informative differences are not in raw competence but in embodiment, energy constraints, continual learning, and the grounding of goals, values, morals, and beliefs. Consciousness remains an open empirical question: current models are competent without any evidence of experience, which sharpens rather than settles the question of what experience is for. Sections 1–8 are each paired with an open-source computational illustration—a simulation, figure, or animation—in an accompanying repository, so that the argument can be read, run, and modified.

**Keywords:** foundation models; large language models; predictive processing; consciousness; neural network architectures; multistable dynamical systems; rational agency; values and alignment; mechanistic interpretability

## Contents

**1 Introduction: a productive comparison** **2**

**2 Foundation models: LLMs and LMMs in brief** **6**
2.1 Tokens, attention, and the next-token objective . . . . . 6
2.2 From a laptop toy to a foundation model . . . . . 9

**3 The brain as a prediction machine** **9**
3.1 The free-energy principle and predictive processing . . . . . 9
3.2 Active inference: prediction that acts . . . . . 10
3.3 Active inference versus reinforcement learning . . . . . 10

**4 Architectures compared: biological and artificial networks** **13**

**5 Thinking and acting, humanly and rationally** **15**
5.1 Acting humanly . . . 15
5.2 Thinking humanly . . . 16
5.3 Thinking rationally . . . 17
5.4 Acting rationally . . . 18

**6 Optimization and multistable dynamics** **18**
6.1 A teaching landscape . . . 18
6.2 Training determines the attractors; attention retrieves . . . 19
6.3 Multistability and perturbation . . . 19
6.4 A control pattern is not a second energy . . . 20

**7 Consciousness: the open frontier** **22**
7.1 Competence without experience . . . 22
7.2 Access, ignition, and a toy workspace . . . 22
7.3 A verbalizable slice is not phenomenal consciousness . . . 23

**8 Values, goals, morals, and beliefs** **24**
8.1 Need versus a statistical regularity . . . 24
8.2 The trolley as next-token mass . . . 25

**9 From comparison to test: a neuro–AI programme** **27**
9.1 How a model computes, versus why an answer looks plausible . . . 27
9.2 Methods, in two directions . . . 28
9.3 From model–brain correspondence to causal tests . . . 29

**10 Conclusion: what remains distinctly human?** **29**

# 1 Introduction: a productive comparison

In the sciences of mind, the brain has often been described by analogy with the leading technology of the period: hydraulics, clockwork, the telegraph, the digital computer. Foundation models change the direction of that comparison. The question is no longer only whether the brain resembles a machine. It is also whether current machines resemble us, and which of their successes and failures constrain claims about human cognition (Russell and Norvig, 2021; Bommasani et al., 2021).

This paper addresses the second question by *subtractive comparison*. Large language models are not brains, and brains are not transformers. The comparison is useful when it is specific. Where the two systems agree, the shared property is a candidate for a computational requirement. Where they differ, the difference is a candidate for a biological or human constraint. Neither inference follows from resemblance alone: a shared property must survive a controlled change of implementation, and a proposed human constraint must explain a capacity or condition absent from the model.

Marr distinguished three levels at which an information-processing system must be described (Marr, 1982): *computational theory* (what is computed, and why that mapping is appropriate); *representation and algorithm* (which internal variables are used, and how they are transformed); and *hardware implementation* (how the representation and algorithm are realized physically). The levels are only loosely coupled: agreement at one does not imply agreement at the others, and the same objective, or the same representation, can be realized by different algorithms on different physical substrates. This paper has two themes, and the separation serves the first. From brains to models uses Marr's levels directly: the descriptions in the sections that follow—function, dynamics, and architecture—are his three rows. The second theme, what AI

reveals about being human, starts from implementation (synapses, spikes, energy, a body) but does not stay there. Viability, phenomenal experience, and values grounded in need are not properties of a hardware description.

Figure 1 draws that table for two substrates. The boxes are teaching labels used in the sections that follow, not recordings from cortex and not an evaluation of a large language model.

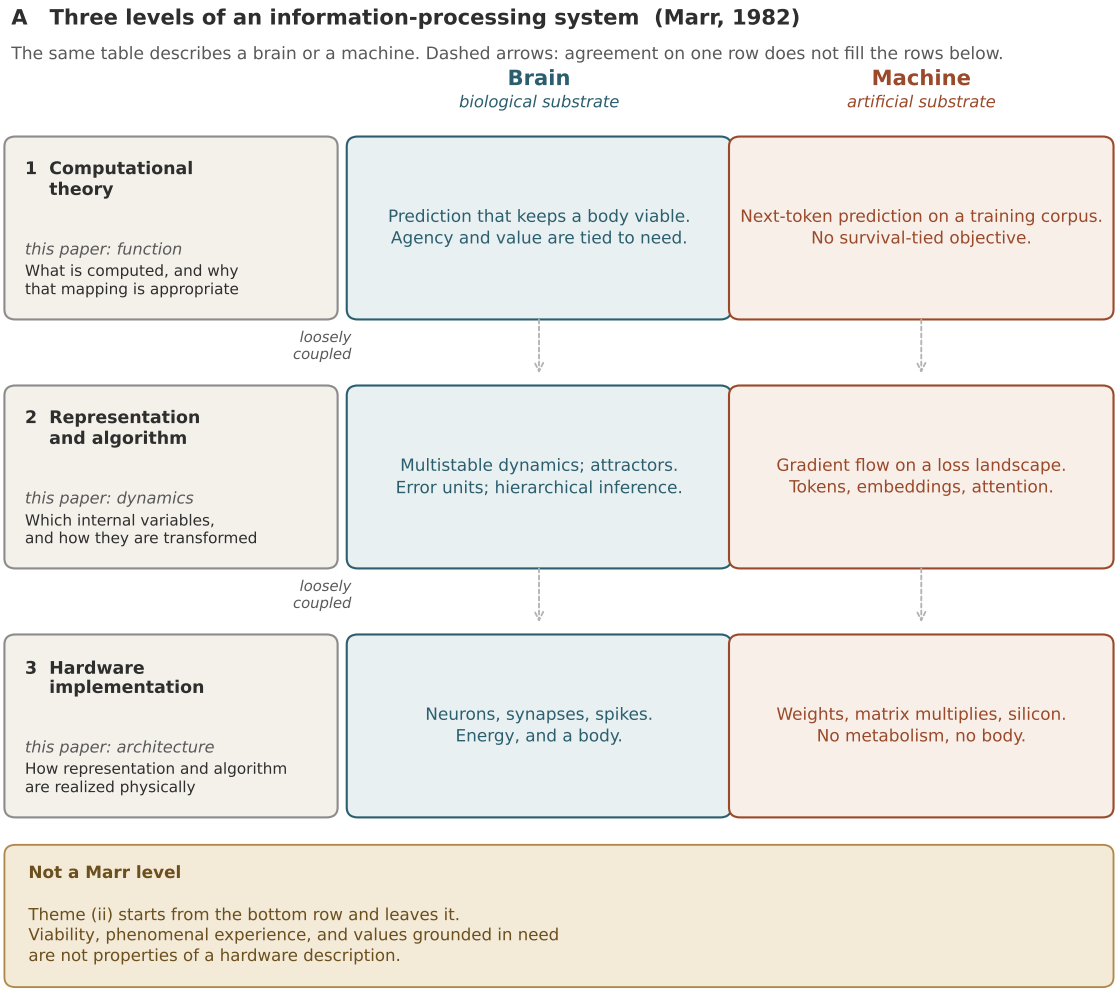



Figure 1: Marr's three levels of description for an information-processing system, drawn for a brain and for a machine (Marr, 1982). Each row is one level: computational theory (what is computed, and why); representation and algorithm (which variables, and how they are transformed); hardware implementation (how that is realized physically). The paper's names—function, dynamics, architecture—are the same three rows. Dashed arrows mark loose coupling: agreement on a row does not imply agreement below it. The band under hardware is not a fourth Marr level; theme (ii) starts from implementation (synapses, spikes, energy, a body) and then leaves it. Generated by `code/notebooks/00_intro_timeline.ipynb`; rebuild with `make figures`.

*Remark:* The usual direction of work in neuroimaging and computational medicine runs from measurements (images, signals, clinical records) to models (atlases, classifiers, predictive markers). Foundation models reverse that order: they arrive already fluent, and the task is to state what they are not.

## How to read this paper

The paper supports two reading paths through one argument, not two separate accounts. Clinicians and neuroscientists can follow the biological question, the observable contrast, and the stated limit of each teaching model. Mathematically inclined readers can follow the objective, state variables, intervention, and failure condition, with the notebooks providing the executable detail.

> **Clinical and neuroscientific reading.** For each comparison in the paper, ask three questions. What was observed—the behaviour, signal, or clinical pattern? What biological interpretation is proposed for it? And which clinical or neural measurement could distinguish that interpretation from a competing one? The answers locate the claim in a body and in the clinic.
>
> **Computational reading.** For the same comparison, identify four things: the objective being optimized, the representation (the state variables used), the dynamics (how those variables are transformed), and the intervention that would test the claim. A similarity between brain and model is informative only at the level—function, dynamics, or architecture—on which it was established.

Three approaches make either path concrete. *Name the level.* Every comparison in the paper is made at one of Marr's three levels (Figure 1), which the sections rename *function* (what is computed, and why: prediction, agency, value), *dynamics* (which variables are used, and how they are transformed: attractors, gradient flow), and *architecture* (how that is realized physically: synapses and spikes versus attention and weights). Because the levels are only loosely coupled, a similarity is informative only at the level on which it was established: agreement on a row is a candidate computational requirement, and disagreement on a row a candidate biological constraint. *Apply the decision rule.* Where brain and model agree, ask whether the shared property survives a change of implementation; where they differ, ask whether the missing biological condition explains a capacity or a constraint. This is subtractive comparison, and resemblance alone passes neither test. *Watch for the remainder.* Embodiment, viability, phenomenal experience, and value grounded in need are not a fourth Marr level—they are the band under hardware in Figure 1—and they are what the second theme isolates once the three levels are accounted for.

Table 1 applies these approaches in advance. For each group of sections it states three things: the comparison being tested, the Marr level at which that comparison holds (function, dynamics, or architecture), and the inference the comparison does *not* license. Because most comparisons with human cognition are drawn at the level of function—where brain and model share only an objective—the entries in the last column typically name the missing remainder (embodiment, viability, phenomenal experience, or grounded value) rather than a competing mechanism at the same level.

Sections 1–8 are each paired with a notebook (notebook $N$ with section $N{+}1$; a $b$ or $c$ suffix is another view of the same section). Sections 9–10 synthesize the comparisons and therefore have no new teaching model. The comparison also has a direction. For most of the history of the sciences of mind it ran one way: the newest machine—hydraulics, clockwork, the telegraph, the digital computer—was used to explain the brain (Figure 2). Foundation models are the first machines fluent enough to make the comparison run in both directions, so that one now asks not only whether the brain resembles a machine but whether the machine has come to resemble us. The timeline is the history of that comparison; Figure 1 is the method by which the later sections read it.

> ▷ ***Explore in code:*** *Marr's three levels and the timeline of brain metaphors (interactive catalogue)*
> `code/notebooks/00_intro_timeline.ipynb`

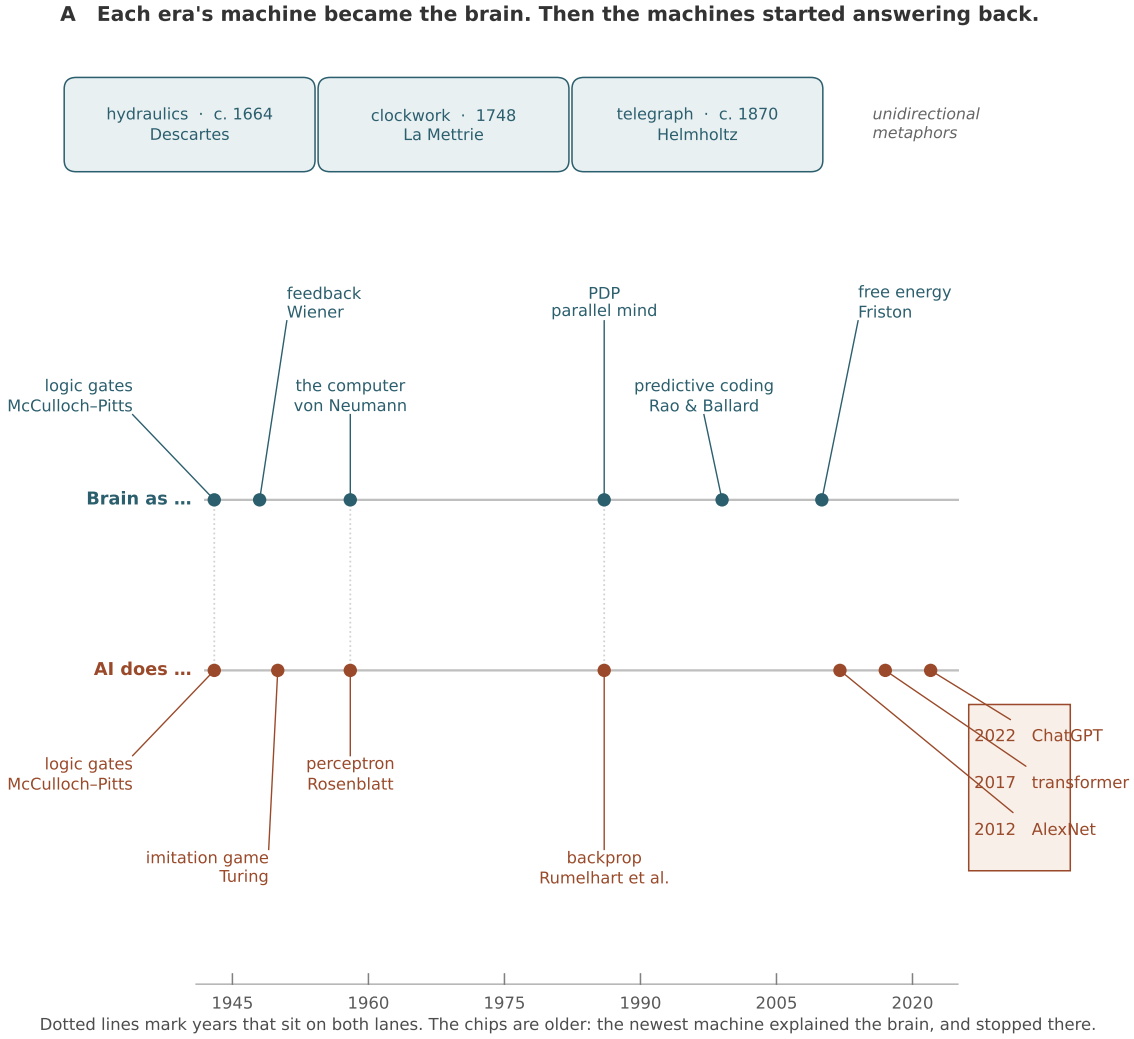



Figure 2: Brain–machine analogies and AI milestones on a common time axis. Before McCulloch–Pitts the comparison is unidirectional: the newest machine is used to explain the brain. After that point the same devices appear in both histories. The comparison becomes bidirectional when machines produce language at a scale that raises the reverse question. Generated by `code/notebooks/00_intro_timeline.ipynb`; rebuild with `make figures`.

Table 1: The argument in two passes. For each group of sections the table names the comparison tested, the Marr level at which it is made (Figure 1: *function* = computational theory, *dynamics* = representation and algorithm, *architecture* = hardware), and what the comparison does not establish. Embodiment, viability, experience, and grounded value are not a Marr level; they are the remainder in the last column.

| Sections | Marr level | Comparison tested | Limit of the inference |
|---|---|---|---|
| 2–3 | Function | Token prediction and biological prediction share an inferential vocabulary; action introduces preferences and information-seeking. | A predictive objective alone does not supply embodiment or viability. |
| 4–6 | Architecture and dynamics | Representations and dynamical descriptions can converge while learning rules, substrates, and timescales diverge. | A shared representation or attractor description does not imply a shared mechanism. |
| 7 | Function | Reportability and flexible access can be operationalized in brains and models. | Access-like organization and self-report do not establish phenomenal experience. |
| 8 | Function | Human judgments can be reproduced as conditional statistics learned from a corpus. | Reproducing a judgment does not establish a value grounded in need. |
| 9–10 | Dynamics | Neural and mechanistic methods turn analogies into predictions and interventions. | Correlation, decodability, or fluency alone does not identify a mechanism or a distinctively human condition. |

## 2 Foundation models: LLMs and LMMs in brief

A foundation model is a large neural network trained on a broad corpus by self-supervised prediction, then adapted to many downstream tasks (Bommasani et al., 2021). Almost all such models today are transformers (Vaswani et al., 2017). A large language model predicts the next token in a stream of text. A large multimodal model applies the same objective to images, audio, video, or action. The architecture and the training signal are shared; only the data type changes.

### 2.1 Tokens, attention, and the next-token objective

A transformer operates on a sequence of discrete symbols called *tokens*. A token is the smallest unit the model reads and predicts: in the toy of Figure 3 it is a single character, whereas production models use sub-word pieces. The fixed inventory of possible tokens is the *vocabulary* $\mathcal{V}$. Each token is mapped to a vector, its *embedding*—a point in a high-dimensional space in which tokens used in similar ways sit close together. Because the network reads all positions at once and is otherwise blind to order, a *position embedding* is added to encode where in the sequence each token lies.

The operation that gives the architecture its name is *attention*: a weighted look-back by which each position summarizes the earlier ones. Each position emits a *query* (what it is looking for), each earlier position a *key* (what it holds) and a *value* (what it passes on); the agreement between a query and a key sets a weight, a *softmax* normalizes the weights so they sum to one, and the position's updated representation is the weighted average of the values. A *causal mask* sets the weight on any not-yet-written token to zero, so prediction stays strictly left-to-right, as in generation. Several such look-backs—*attention heads*—run in parallel, each free to track a different regularity (Vaswani et al., 2017). A position-wise feed-forward network then refines each position on its own, and a final linear map followed by a softmax returns a probability

distribution over the next token.

Training adjusts the weights to minimize the *cross-entropy*, or negative log-likelihood, of the token that actually followed: the score is low when the model placed high probability on the observed next symbol,

$$\mathcal{L} = -\log p_\theta(x_{t+1} \mid x_{\le t}).$$

A model that spread its mass uniformly over the $|\mathcal{V}|$ tokens would score $\log|\mathcal{V}|$ (here $\ln 27 \approx 3.3$ nats), and *gradient descent* by *backpropagation* drives $\mathcal{L}$ below that chance value. The trained model is then a family of conditional distributions $p_\theta(x_{t+1} \mid x_{\le t})$ and nothing more: fluency is a by-product of a low score, not a separate objective for meaning.

Three facts about these models set up the comparison that follows. First, the training objective is simple: predict the next token. From that single objective, trained at scale, comes competence that transfers to tasks the model was never shown directly. Second, this competence appears as the model grows, by a route that is still only partly understood. Third, the model is disembodied: it acquires human knowledge through text and images, not through a body that acts in a world.

In clinical use, the same objective can appear as fluent completion of a note, report, or explanation. The completion may be useful, but fluency does not show that the model has represented the patient's condition, checked the claim against the world, or adopted the patient's outcome as a goal. This is the first subtractive comparison: next-token competence survives without clinical responsibility or a body that bears the consequence.

Figure 3 shows the objective at a scale that runs on a laptop. A one-layer, four-head, character-level transformer is trained on a small designed corpus in which the same nouns recur across a story, a headline, an encyclopedia line, and a FAQ. Each attention head is a weighted look-back over the prompt: a query character (row) attends to earlier characters (column), and a causal mask forbids attention to tokens that have not yet been written. After a few hundred training steps the heads specialize—some stay local, others reach back to the shared sentence frame—and next-character probability mass concentrates on the letters that begin the objects the corpus uses. A production model differs in scale, not in kind: the same objective and the same attention, over a far larger corpus, with embodiment absent from the training signal.

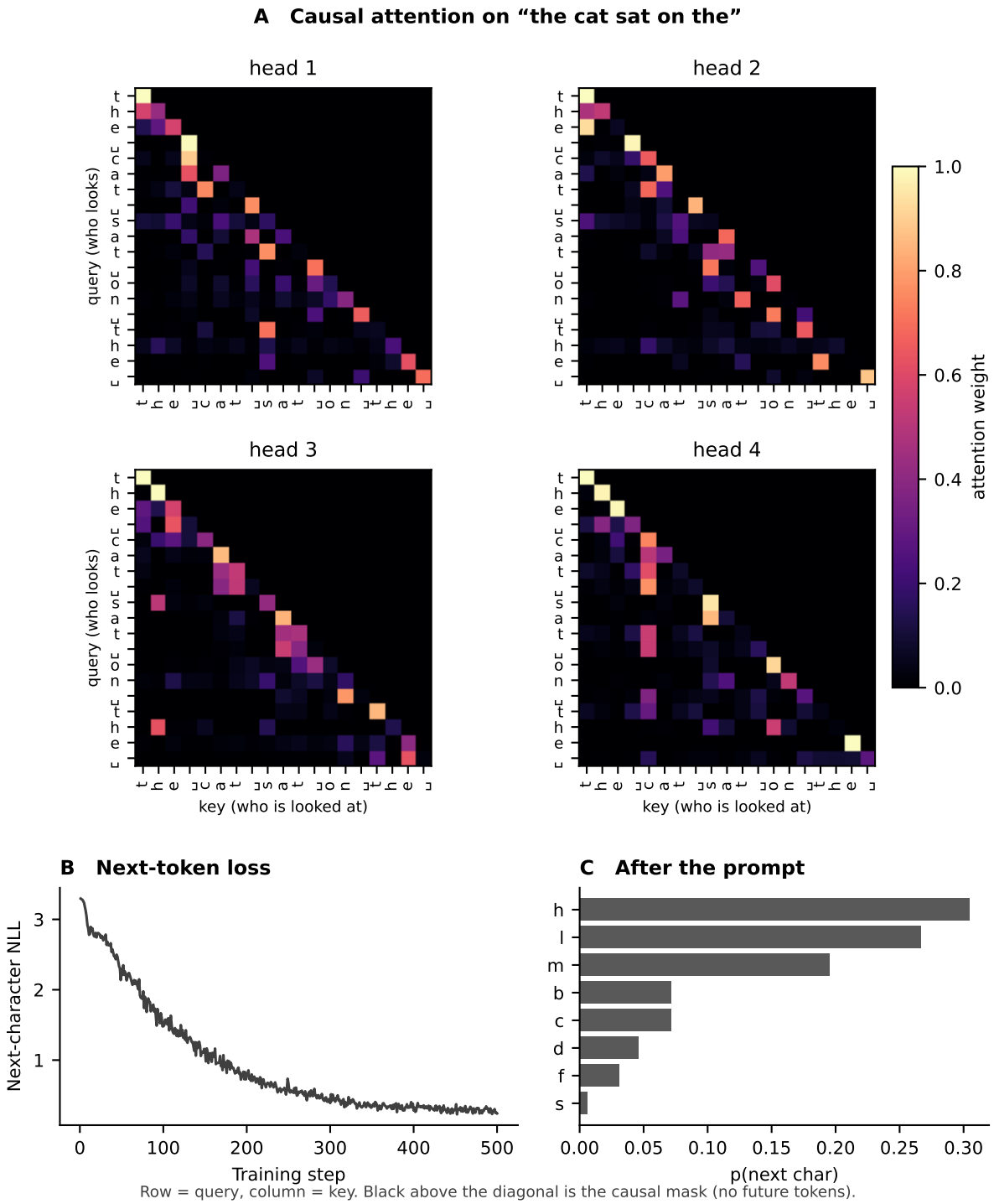



Figure 3: A one-layer, four-head character-level transformer trained on a small designed corpus (story, headline, encyclopedia line, FAQ). (A) Causal attention for the prompt `the cat sat on the`. Each cell is the weight a query character (row) places on a key character (column); the black upper triangle is the causal mask. Heads differ: some stay local, others reach back to earlier words. (B) Next-character negative log-likelihood, which falls from the chance value $\ln 27 \approx 3.3$ as the shared sentence frame is learned. (C) After the prompt, probability mass concentrates on the letters that begin the objects in the corpus (`h`/`l`/`m`/`b`), with a remainder on animal-initial letters. Generated by `code/notebooks/01_tiny_transformer.ipynb`; rebuild with `make figures`.

### 2.2 From a laptop toy to a foundation model

At scale the objective gains a property the toy does not show. A large model can adopt a new format, a translation pair, or a simple rule from a few examples placed in its *context window*—the span of tokens it can read at once—with no change to its weights; this is *in-context learning* (Brown et al., 2020). Instruction tuning and preference optimization then shape the predictor into a conversational interface (Ouyang et al., 2022). What resembles agency in that interface is the same conditional distribution $p(\text{next} \mid \text{past})$, now steered by human feedback; no body has been added. A production model therefore differs from the toy in the size of $\mathcal{V}$, the number of layers, and the corpus, not in the objective or the attention it uses. Section 3.3 turns on this point: a preferred token is not a preferred observation for an agent that must keep itself viable.

> **Clinical and neuroscientific reading.** A foundation model completes text by estimating which token most plausibly comes next; its fluency reflects probability concentrated on a string, not an examination of a patient or a check against the world.
>
> **Computational reading.** The model is a family of conditional distributions $p_\theta(x_{t+1} \mid x_{\le t})$ fit by minimizing cross-entropy. Attention supplies a causally masked, data-dependent average over past representations, and scale changes $\theta$ and the corpus, not the objective.
>
> ▷ ***Explore in code:*** *A minimal transformer trained on character-level text; attention maps* `code/notebooks/01_tiny_transformer.ipynb`

## 3 The brain as a prediction machine

Machine learning and theoretical neuroscience now share an inferential vocabulary, not a demonstrated common objective or mechanism. In AI, next-token prediction proved sufficient to absorb much of human knowledge. In neuroscience, prediction came to organize accounts of perception and action. Predictive processing describes perception as controlled inference: the cortex holds a generative model of the world and updates it by reducing prediction error (Rao and Ballard, 1999; Friston, 2010; Clark, 2013; Hohwy, 2013).

### 3.1 The free-energy principle and predictive processing

The free-energy principle is the most general form of this view. In that framework, a system that persists in its environment is described as minimizing a bound on the surprisal of its sensory states, a quantity called variational free energy (Friston, 2010). Perception then becomes approximate Bayesian inference. Hierarchical cortical circuits pass predictions downward and prediction errors upward, and precision weighting plays the role of attention (Rao and Ballard, 1999; Clark, 2013; Hohwy, 2013). A single imperative—reduce prediction error—thus covers perception, attention, and learning, in a mathematical vocabulary (variational inference, generative models) that machine learning shares.

For observed sensory states $s$, latent causes $z$, and an approximate posterior $q(z)$, the relevant bound can be written

$$F[q] = \mathbb{E}_{q(z)}\big[\log q(z) - \log p(s, z)\big] \geq -\log p(s).$$

The clinical reading is that perception depends on both incoming evidence and a model of its causes; the computational reading is that inference changes $q$ to tighten the bound. The predictive-coding notebook implements the narrower error-minimization mechanism, not the general claim that every persistent biological system performs this particular update.

### 3.2 Active inference: prediction that acts

Active inference closes the loop. An agent can reduce prediction error in two ways: by updating its model, or by *acting on the world* so that its predictions come true (Friston et al., 2017; Parr et al., 2022). Action selection becomes inference over policies, scored by *expected free energy*. That score has two terms: a pragmatic term for realizing preferred observations, and an epistemic term for resolving uncertainty about hidden states. Goals are not supplied as an external reward. They are encoded as *prior preferences* over observations: a fish expects to be in water, and acts to fulfil that expectation.

At the grain of the T-maze demonstration, this can be read schematically as

$$G(\pi) = \text{expected preference cost} - \text{expected information gain}.$$

Sign conventions differ across formulations; the operational prediction does not. When the cue is informative, removing the information-gain term should remove cue-seeking unless another exploration rule replaces it. That intervention is the mathematical claim made runnable in notebook 02b.

### 3.3 Active inference versus reinforcement learning

Reinforcement learning (RL) makes a useful contrast (Friston et al., 2009; Sajid et al., 2021). An RL agent maximizes expected cumulative reward, an external scalar, usually by learning a value function or a policy, and exploration is added by hand ($\varepsilon$-greedy choices, entropy bonuses). An active-inference agent minimizes expected free energy, and the balance between exploration and exploitation follows from that objective: the epistemic term values information gain, the pragmatic term values preference satisfaction, and their relative weight shifts with the agent's uncertainty. Three further differences follow. Active inference is model-based by construction, with beliefs over hidden states and planning cast as inference. Reward is replaced by prior preferences, so value is a property of beliefs rather than of the world. Perception and action minimize one objective rather than being handled by separate machinery. The two frameworks are formally close: RL is recovered when preferences encode reward and the epistemic term is dropped, and both belong to the control-as-inference literature (Levine, 2018). The difference that matters here is conceptual. In active inference an agent's goals are expectations about its own existence, which is closer to how biological value is grounded (Section 8).

Table 2: Reinforcement learning versus active inference.

| | Reinforcement learning | Active inference |
|---|---|---|
| Objective | maximize expected reward | minimize expected free energy |
| Source of goals | external scalar reward | prior preferences over observations |
| Exploration | added ($\varepsilon$-greedy, bonuses) | epistemic value, emergent |
| World model | optional | required (generative model) |
| Scale today | large | small discrete systems |

The bridge across the section is now explicit. Predictive processing is the framework in which perception depends on predictions and errors; predictive coding is the narrower circuit mechanism illustrated in Figure 5. Variational free energy supplies an inference objective, and active inference extends the comparison from latent-state estimation to policy selection. The T-maze then turns one term of expected free energy into a removable component: delete expected information gain and cue-seeking should disappear unless exploration is supplied elsewhere.

Figure 4 places both agents in one T-maze. A hidden context puts the reward in the left or the right arm, and a cue location reveals which arm without risk. An active-inference agent

with a two-step policy horizon visits the cue on the first trial—expected free energy pays for the information—and then takes the baited arm. An $\varepsilon$-greedy Q-learner, which sees only a scalar reward, wanders. The gap persists across forty trials: cue-first behaviour and return are at ceiling for active inference from trial 1, while Q-learning is still sampling. The notebook is a NumPy stand-in for the same task as the `pymdp` demo (Heins et al., 2022).

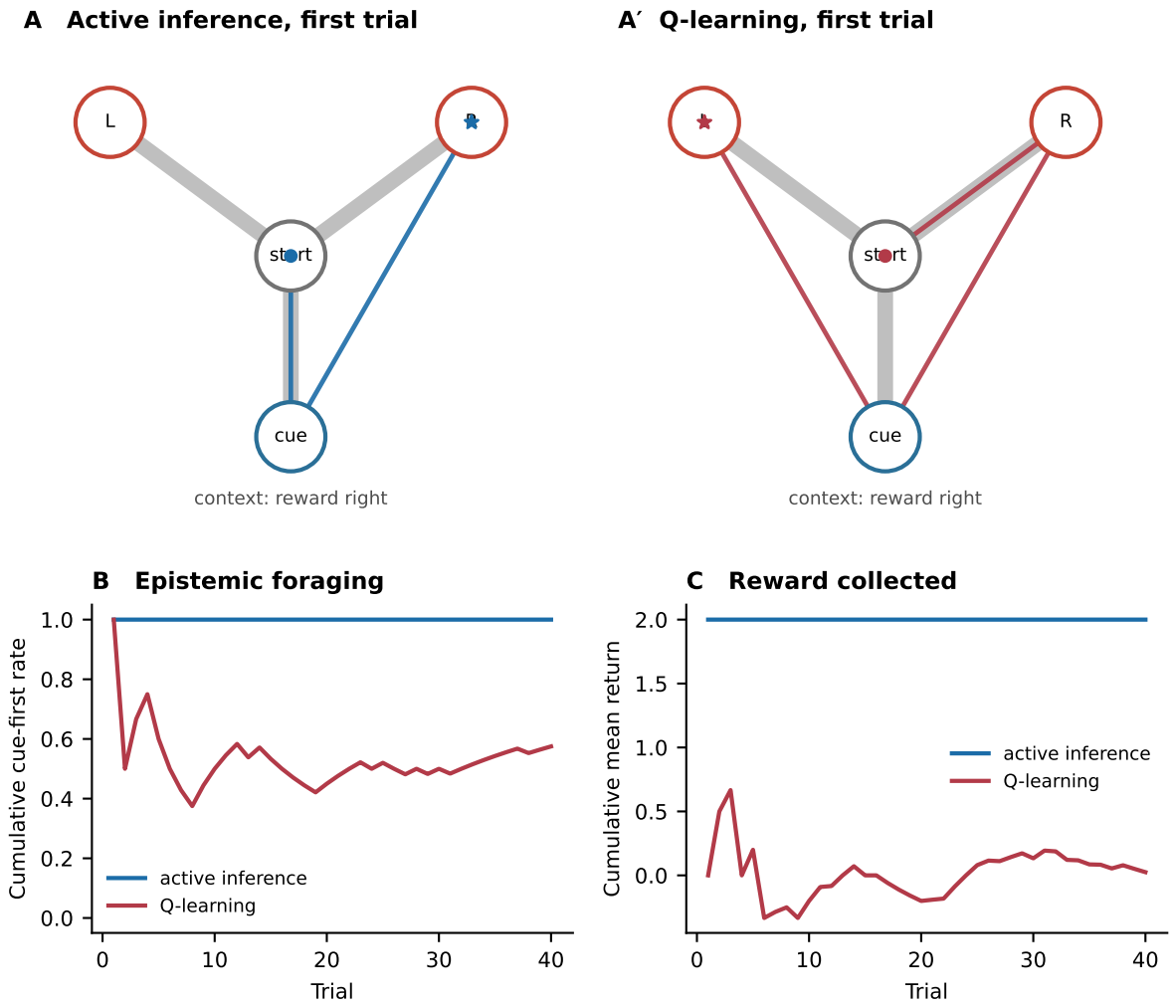



Figure 4: Active inference versus Q-learning in a T-maze. (A) First trial under active inference: start → cue → baited arm. (A′) First trial under $\varepsilon$-greedy Q-learning. (B) Cumulative rate of visiting the cue before an arm. (C) Cumulative mean trial return. Generated by `code/notebooks/02b_active_inference_vs_rl.ipynb`; rebuild with `make figures`.

On this view the similarity between brains and LLMs is not architectural but lies in the objective: both are prediction machines. The difference is what is predicted, and why. The brain predicts the sensory consequences of its own actions, in a body that must keep itself far from thermodynamic equilibrium. An LLM predicts tokens in a corpus from which viability constraints have been removed. Allostasis, interoception, and affect give biological prediction its urgency, and perhaps its felt quality (Seth, 2021).

Figure 5 makes the cortical side of that claim concrete. A two-level network in the style of Rao and Ballard (Rao and Ballard, 1999) is trained on $8 \times 8$ patches from synthetic $1/f$ images mixed with a few windowed gratings—enough oriented structure for a laptop-scale dictionary to find. The first-level basis functions become localized and oriented. Iterated inference then

explains away a held-out patch, driving the residual toward zero. The notebook is the runnable version of the same figure.

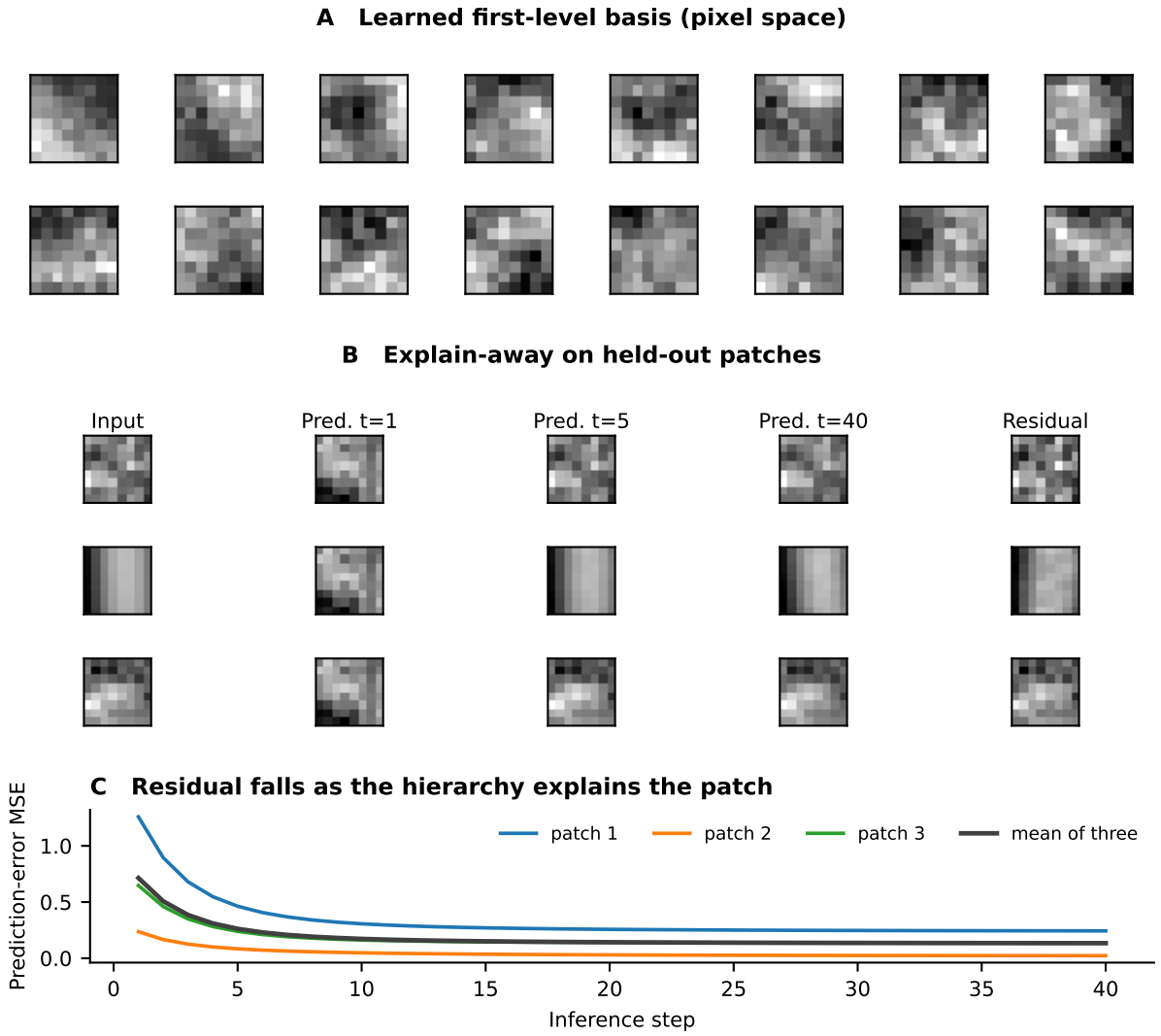



Figure 5: Two-level predictive coding on $8 \times 8$ patches from $1/f$ images mixed with windowed gratings. (A) Learned first-level bases, shown in pixel space after inverting ZCA whitening. (B) Three held-out patches: input, reconstructions at inference steps 1, 5 and 40, and the final residual. (C) Prediction-error MSE in the whitened space the network sees. Generated by `code/notebooks/02_predictive_coding.ipynb`; rebuild with `make figures`.

▷ ***Explore in code:*** *Predictive coding network (Rao–Ballard style) learning $1/f$ image patches; hierarchical error minimization*
*`code/notebooks/02_predictive_coding.ipynb`*

▷ ***Explore in code:*** *NumPy expected-free-energy agent in a T-maze, side by side with a Q-learning agent: epistemic foraging versus $\varepsilon$-greedy exploration*
*`code/notebooks/02b_active_inference_vs_rl.ipynb`*

**Clinical and neuroscientific reading.** Prediction in a clinical or neural system is a hypothesis about the causes of observations; action adds consequences and information seeking. The figures isolate those claims but do not identify a cortical implementation.

**Computational reading.** Notebook 02 changes prediction error during inference; notebook 02b changes the epistemic term in policy scoring. These are two different objectives

and two different interventions.

The limits follow the same separation. The image-patch network is not a cortex, the discrete T-maze is not an organism, and its prior preferences are designed numbers rather than measured allostatic needs. The demonstrations show how prediction, inference, and information-seeking can be related formally; they do not establish that biological persistence uses the particular updates implemented here.

## 4 Architectures compared: biological and artificial networks

Biological neural networks are recurrent, spiking, neuromodulated, sparsely active, and locally plastic. They learn continually, on a budget of about 20 watts, without catastrophic forgetting. Transformers, at inference, are largely feedforward and dense; they are trained once by global gradient descent and then frozen. Backpropagation in its standard form is biologically implausible, though approximations exist (Lillicrap et al., 2020; Richards et al., 2019). Despite these differences, the representations learned by large models predict cortical activity during language and vision with substantial fidelity (Schrimpf et al., 2021; Goldstein et al., 2022). Representations can therefore converge even where mechanisms diverge. Table 3 collects the contrasts that the rest of the paper uses.

Table 3: Biological networks versus foundation models. The last row is the one Figure 6 demonstrates: whether an error at the output can change a synapse that did not itself produce that output.

| Dimension | Brain | Foundation model |
|---|---|---|
| Elementary unit | Spiking neuron, analogue dendrite | Real-valued unit; attention head |
| Connectivity | Sparse, recurrent, locally structured | Dense layers; feedforward attention at inference |
| Learning rule | Local plasticity (Hebb, neuromodulated STDP); no known exact backprop | Global gradient descent (backprop), mostly in batch |
| Timescales | ms spikes; minutes-to-years plasticity; development | One training run, then a frozen forward pass (plus a short context window) |
| Energy | $\sim$20 W, continuous | Training at cluster scale (orders of magnitude above 20 W); inference still far above a cortex |
| Protocol | Continual, online, no clean epochs | I.i.d. batches; catastrophic forgetting if naïvely sequential |
| Neuromodulation | Global and local gain (DA, ACh, ...) | A learning-rate scalar, if that |
| Sparsity | Few spikes, few active synapses | Dense activations; sparsity is an add-on |
| Memory | Synapses + complementary episodic systems | Weights + context window |
| Embodiment | Yes (allostasis, action, a body that can fail) | No (text / image / tool interface) |

For neuroscience and clinical modelling, the practical implication is that a representation can predict neural activity or clinical labels while its learning mechanism remains unlike biological learning. A successful encoding model therefore supports a claim about representational correspondence at a stated task and measurement, not a claim that the model learned as the patient or cortex learned. Figure 6 makes that distinction testable by holding the inputs and task fixed while changing credit assignment.

Figure 6 turns the learning-rule row into a toy. Two agents see the same two-dimensional points. A leaky integrate-and-fire (LIF) perceptron encodes each coordinate as a Poisson spike train and updates each synapse from its own pre- and post-synaptic counts, a local delta rule. A two-layer perceptron (MLP) on the raw coordinates assigns error through a hidden layer by stochastic gradient descent (SGD). On a linearly separable pair of clouds, both reach ceiling. On a four-blob XOR, the LIF network stays at chance: it has no hidden representation for the label to reach, and a local rule on the sensors cannot represent exclusive-or. The MLP separates the quadrants in a few epochs. The data are the same; the difference is whether credit assignment is local. This is not a claim that cortex cannot compute XOR. It is a claim that *if* the brain computes it, it does not do so by the perceptron implemented here, and that backpropagation, whatever its biological status, does something extra.

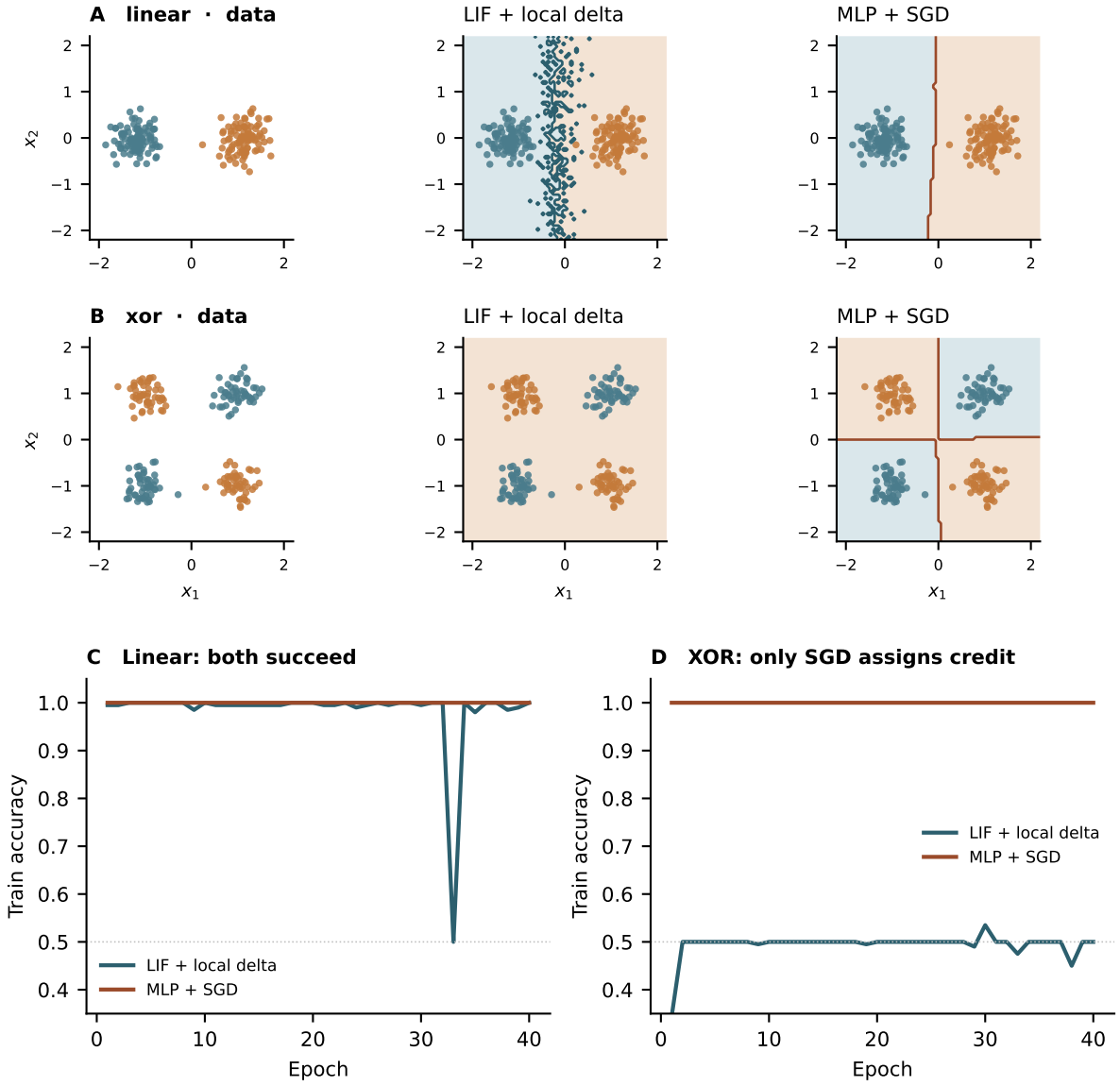



Figure 6: Local spike-count learning versus SGD on the same points. (A) A linearly separable pair: LIF + local delta and a small MLP both separate the clouds (the LIF boundary is noisier because the encoding is Poisson). (B) Four-blob XOR: the local rule stays at chance; SGD assigns credit through a hidden layer. (C–D) Train accuracy over epochs. Generated by `code/notebooks/03_bio_vs_artificial.ipynb`; rebuild with `make figures`.

▷ ***Explore in code:*** *Side-by-side simulation: local spike-count learning in a spiking network versus SGD in a small MLP on the same task*
*code/notebooks/03_bio_vs_artificial.ipynb*

# 5 Thinking and acting, humanly and rationally

Russell and Norvig give four definitions of AI: thinking humanly, thinking rationally, acting humanly, and acting rationally (Russell and Norvig, 2021). These were long treated as a teaching device. Foundation models occupy all four quadrants at once, and each quadrant returns a different verdict on the human comparison.

Figure 7 is the map. The rows ask whether we score the process or the behaviour; the columns ask whether the standard is how people actually perform or how an ideal agent would act. A foundation model can be placed in every cell at once. The subsections below take the cells in turn. Notebook 04 is a controlled demonstration of the upper-left cell only; notebook 04b is the map itself.

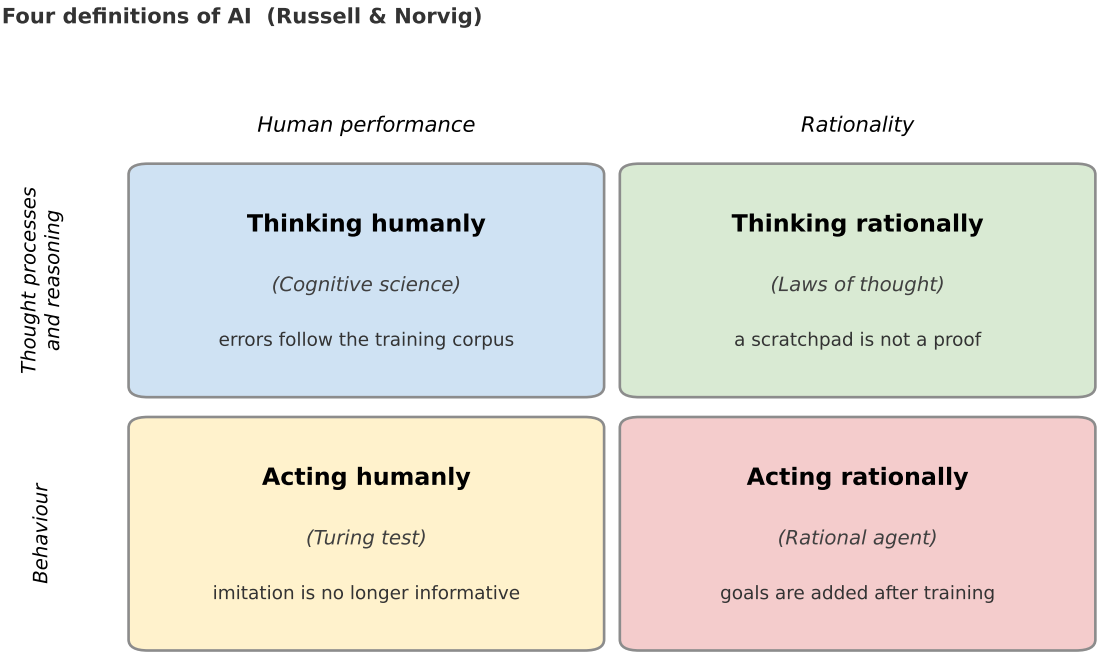



Figure 7: Four definitions of AI, after Russell and Norvig (Russell and Norvig, 2021). Rows score thought versus behaviour; columns compare against human performance versus an ideal of rationality. The third line in each cell is this paper's verdict for foundation models, not AIMA's. Generated by `code/notebooks/04b_four_views_of_AI.ipynb`; rebuild with `make figures`.

## 5.1 Acting humanly

Turing proposed the imitation game in place of the question "can machines think?" (Turing, 1950). If a judge cannot tell the machine from a person by verbal exchange alone, the question is, for practical purposes, answered. Large language models now pass increasingly demanding versions of that game. Behavioural indistinguishability is no longer the informative test.

Imitation establishes competence at the public interface of thought: the strings a person might produce. It does not establish world-involving belief, a body subject to viability constraints, or phenomenal experience. A system trained to continue human text can play the game because

the game *is* text. Figure 3 showed the mechanism; scale and instruction tuning change the interface, not the objective. The later sections on consciousness and values turn on this gap: fluent self-report is expected once self-report is in the training corpus.

### 5.2 Thinking humanly

A stricter test than imitation asks whether a model reproduces not only human answers but human *errors*: framing effects, content effects in logic, the heuristics-and-biases catalogue (Kahneman, 2011). Binz and Schulz found that GPT-3 reproduces several of these signatures and fails others (Binz and Schulz, 2023). Where the match holds, training on human data is the likely cause. Where it fails, the mismatch is consistent with a system that has neither our evolutionary history nor a developmental trajectory.

Figure 8 shows this claim at toy scale. The Asian-disease problem (Tversky and Kahneman, 1981) states one actuarial fact in two wordings: 200 of 600 saved is 400 of 600 dead. People take the sure option under the lives-saved wording ($\approx 72\%$) and refuse it under the lives-lost wording ($\approx 22\%$). A word-level tiny transformer, the same architecture as in Figure 3, is trained on a small designed corpus of framed items. When every gain-frame sentence continues with "the sure option" and every loss-frame sentence with "the risky option," next-token mass after a held-out probe copies the pattern (panels B–C). The same architecture, trained on a corpus that names the sure and risky options equally often in both frames, sits on the chance line (panel D). The bias here is not an evolved heuristic. It is $p(\text{next} \mid \text{past})$ following a corpus that already contained the bias. This does not claim that GPT-3 is a 32-dimensional transformer. It claims that one mechanism sufficient to reproduce a published human error is that the text contained the error. Binz and Schulz are the large-model counterpart; this figure is the controlled demonstration.

The clinical analogue is wording sensitivity in risk communication, consent, triage, or a generated recommendation. If two descriptions preserve the clinically relevant quantities but change the model's answer, the difference is evidence about the decision procedure, not about the patient. The balanced-corpus control asks the narrower causal question: is the observed framing effect sustained by the conditional statistics in the training text?

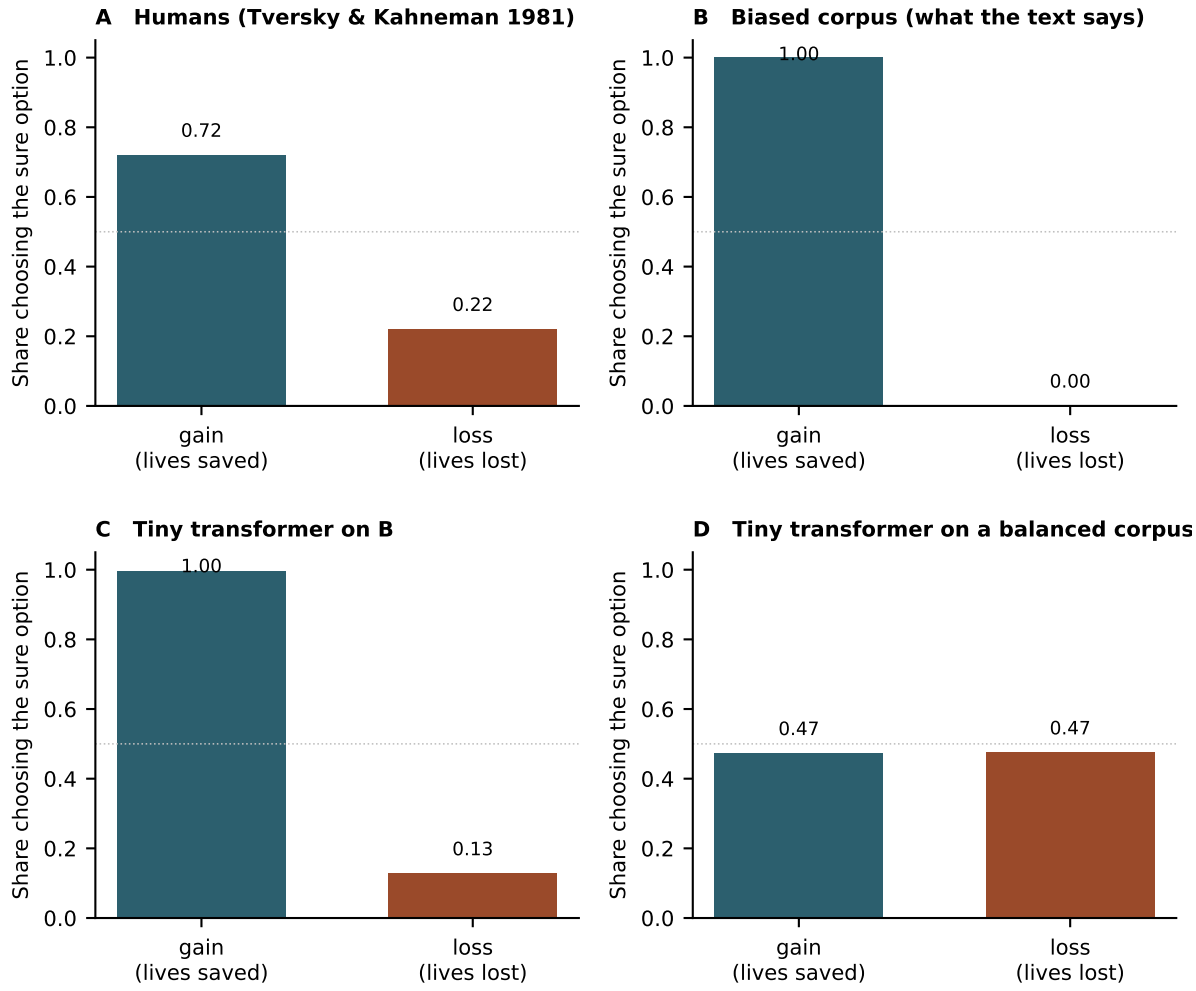



Figure 8: Framing as next-token mass, not as an evolved heuristic. (A) Human share choosing the sure option in the Asian-disease problem (Tversky and Kahneman, 1981). (B) A small designed corpus written so that every gain-frame sentence continues with `sure` and every loss-frame sentence with `risky`. (C) A word-level tiny transformer trained on B copies the pattern after a held-out probe. (D) The same architecture on a corpus that names both options equally often in both frames does not. No open-weights LLM was run. Generated by `code/notebooks/04_quadrants_biases.ipynb`; rebuild with `make figures`.

### 5.3 Thinking rationally

The laws-of-thought tradition asks for valid inference, not a convincing imitation of a person. Chain-of-thought prompting and formal-reasoning benchmarks can look like a System-2 overlay on a System-1 predictor (Kahneman, 2011). The overlay is still next-token prediction with a longer prefix. A scratchpad is not a proof system: the model is constrained by which strings follow "therefore" in the training corpus, not by validity. Hallucination describes the same fact from the other side: assertion without a world that can refute it.

System-1 and System-2 form a useful psychology of *humans*. Applied to a language model, the pair is a metaphor for short versus long samples, not two systems. The model has no fast and slow pair with separate computational budgets and a shared body. It has one objective and a context window. Brittleness under slight rewording is what a predictor whose reasoning is a continuation would be expected to show.

### 5.4 Acting rationally

The last quadrant is the agent view: choose the action that maximizes expected utility given goals. Foundation models arrive with competence and without goals of their own. Goals are imposed afterwards, by the prompt and by preference fine-tuning (Ouyang et al., 2022). This order—capability first, agency added after training—is the reverse of biological development, in which a body subject to viability constraints is present before language.

Section 3.3 drew the distinction already: a preferred token is not a preferred observation in a world the agent must survive. A model that can plan a trip still has no trip it needs to take. Acting rationally, in the textbook sense, requires a utility that belongs to the agent, not one borrowed from the user's instruction and the rater's preferences.

In a clinical workflow, the stakes belong to the patient and the responsibility belongs to people and institutions even when a model proposes the plan. Supplying a utility through a prompt can make an output goal-directed at the interface; it does not make the patient's viability a goal of the model.

> ▷ ***Explore in code:*** *The Russell–Norvig 2×2 as a drawable map, with a lookup for each cell*
> *code/notebooks/04b_four_views_of_AI.ipynb*

> ▷ ***Explore in code:*** *Framing as next-token mass: a tiny transformer on a designed corpus of Asian-disease items, against the human numbers*
> *code/notebooks/04_quadrants_biases.ipynb*

## 6 Optimization and multistable dynamics

Both systems can be described as dynamical systems, but of different kinds. Training a network is a gradient flow on a high-dimensional loss landscape. Cognition unfolds as trajectories through the state space of a recurrent, multistable system. Attractor dynamics—from Hopfield's associative memories (Hopfield, 1982) to winnerless competition and metastable itinerancy (Rabinovich et al., 2008; Kelso, 1995)—give a language in which perception (for example, bistable figures such as the Necker cube), memory recall, decision-making, and even mood can be described as transitions between basins of attraction.

For clinicians and neuroscientists, the useful claim is at the level of dynamics: an observed trajectory may settle, switch after perturbation, or remain metastable. Calling such a trajectory an attractor does not identify its biological substrate, and a two-dimensional teaching landscape is not a measured disease state. It specifies which perturbation and recovery pattern would have to be observed for the description to be useful.

### 6.1 A teaching landscape

Figure 9 is a schematic of that language. A surface $U(\mathbf{X})$ over a two-dimensional slice of activity space has more than one valley. Gradient descent is a path into a well, and the well is an attractor: a memory, a percept, a decision (Kandel et al., 2013). The drawing is a teaching slice, not a cortex and not $5 \cdot 10^9$ neurons. Notebook 05 is a controlled demonstration of the same idea: a Hopfield energy that cannot rise, a Necker two-well that can switch, and attention as one modern-Hopfield step. An animation of the descent is in the companion repository (`code/animations/energy_landscape.gif`).

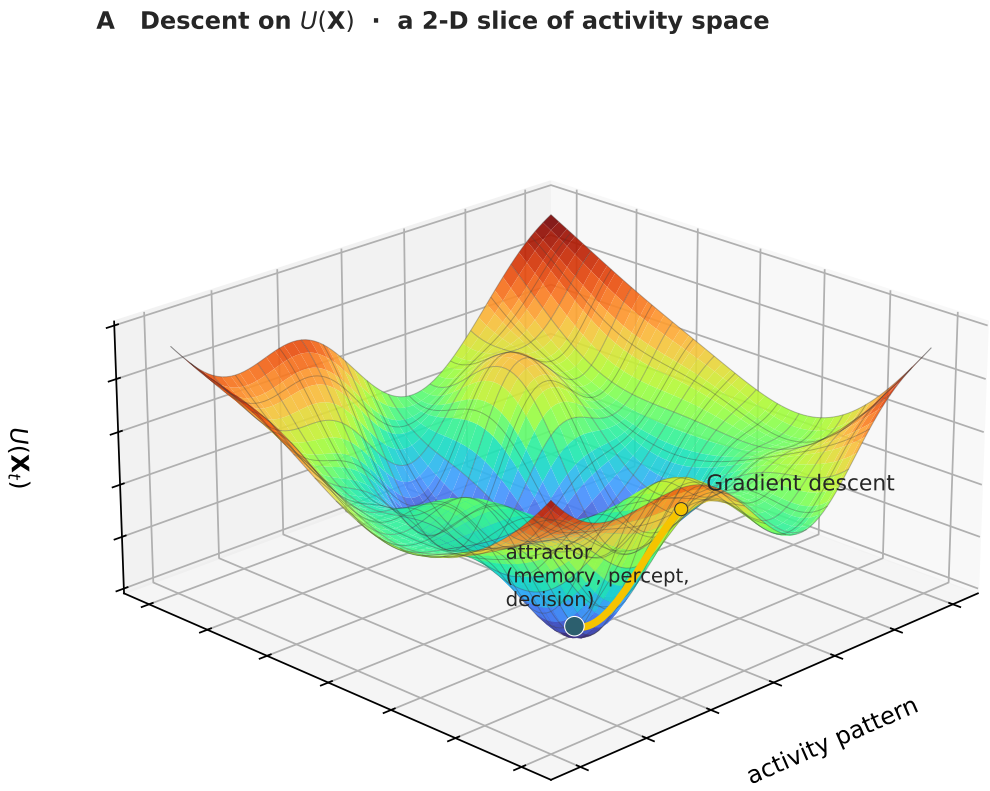



Figure 9: Descent on a multi-well energy $U(\mathbf{X}_t)$, after the schematic in Kandel et al. (2013). The plane is a two-dimensional slice of activity space; the yellow path is gradient descent run to a local minimum (teal). The surface is a designed teaching function, not a measured cortical energy and not the Hopfield $E$ of Figure 10. Generated by `code/notebooks/05b_multistable_animation.ipynb`; rebuild with `make figures`. Animation: `code/animations/energy_landscape.gif`, `.mp4`.

### 6.2 Training determines the attractors; attention retrieves

The two descriptions are not competitors. Training determines the landscape that inference later occupies. A classical Hopfield network stores patterns as minima of $E = -\frac{1}{2} s^\top W s$; recall is descent, and the energy cannot rise (Hopfield, 1982). A transformer is not trained by descending that $E$. But a modern Hopfield update *is* attention: one softmax over similarities to stored keys (Ramsauer et al., 2020). In that sense the attention maps of Figure 3 are a retrieval step in a landscape set by the training corpus. The loss that shaped the keys is not the energy that retrieves them.

### 6.3 Multistability and perturbation

Multistability is the other half. An ambiguous figure is two wells with enough noise to switch between them. An ambiguous prompt is not a Necker cube, but mode-switching in a language model—the same prefix completed one way or the other—is analogous at the level of basins: two attractors, a shared start, and a slight rewording or a sample as the perturbation. Winnerless

competition and metastable itinerancy (Rabinovich et al., 2008; Kelso, 1995) are the regime in which the system does not settle; sequential thought lives there more than a single recall does. Figure 10 shows settling and switching, not itinerancy. It does not claim that a foundation model has a Necker cube. It claims that one shared description of both facts is attractors and perturbations.

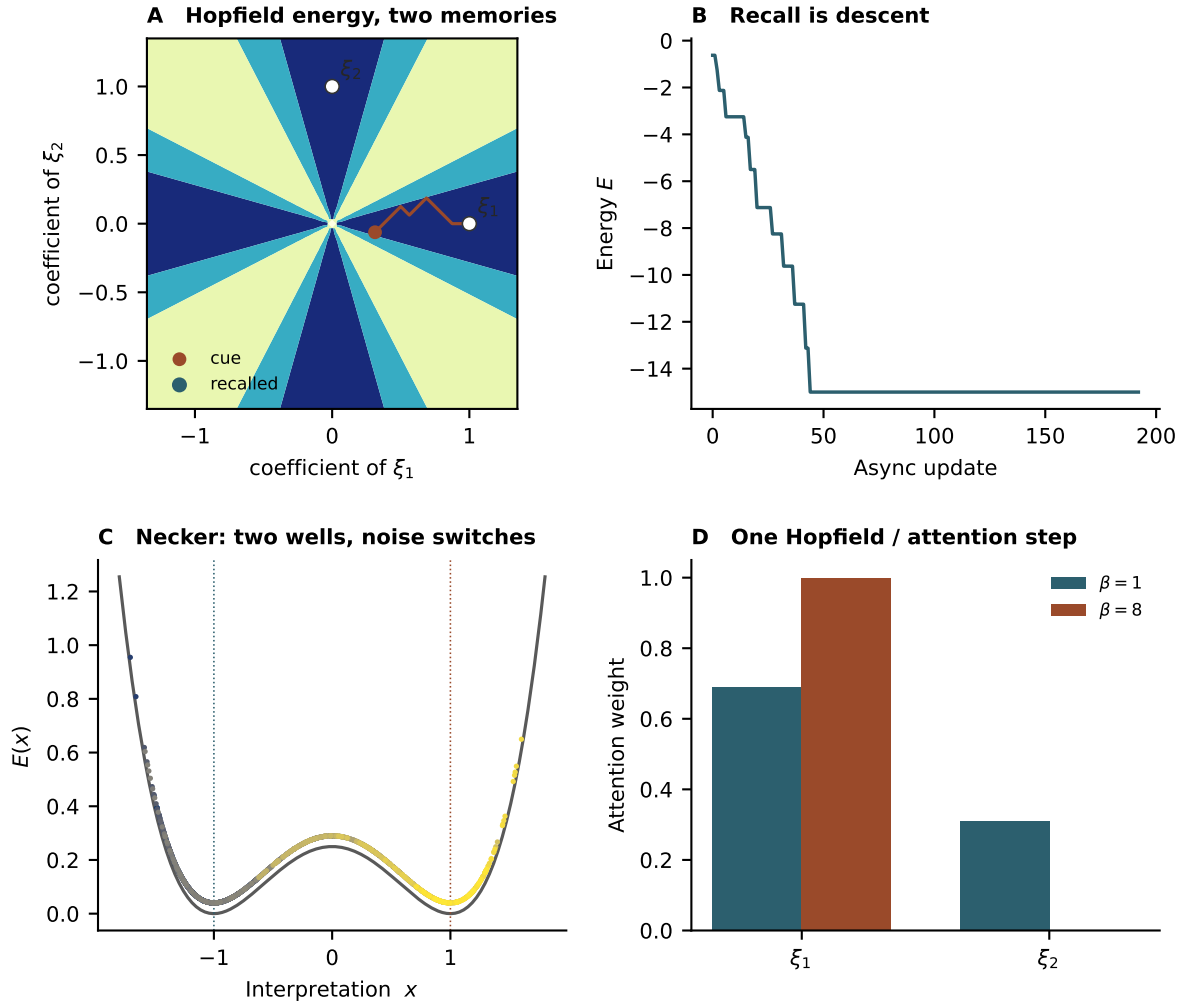



Figure 10: Basins, descent, switching, and one attention step. (A) Energy of a 32-unit Hopfield net in the plane of two stored patterns $\xi_1, \xi_2$. A 35%-corrupted cue (rust) descends into $\xi_1$. (B) The same recall as a monotone drop in $E$. (C) A double-well $E(x) = \frac{1}{4}(x^2 - 1)^2$ as a Necker cube: wells are the two interpretations; Langevin noise switches between them (points coloured by time). (D) One modern-Hopfield / attention step $\xi \leftarrow X\,\mathrm{softmax}(\beta X^\top \xi)$ on a query slightly closer to $\xi_1$. Soft $\beta$ splits the mass; sharp $\beta$ retrieves one key. Generated by `code/notebooks/05_multistable_dynamics.ipynb`; rebuild with `make figures`.

## 6.4 A control pattern is not a second energy

Notebooks 05 and 05b describe settling and switching in an energy. Miller et al. (2026) describe a different dynamical operation on the same Marr row: an alpha/beta spatial *stencil* that suppresses feedforward (gamma-like) activity in some locations and spares others, so that the same multifunctional units change selectivity with context. The control pattern relocates which units are allowed to be active; it does not update a weight matrix. Superposition of two such patterns is analog combination of two contexts (for example, sequential order and report mode),

not a learned map. A traveling open window is then spatial computing in motion: population activity is pulled from a sensory locus toward a memory locus while the drive, and the weights, stay put.

Figure 11 is that claim at teaching grain. A designed 1-D sheet, two object drives, and two stencils produce a unit that keeps an object preference across contexts and a unit that expresses the other object only when the stencil opens that side (panel A). Pairwise means of two control waves yield four distinct spatial gates (panel B). A traveling window relocates expressed drive (panel C). The figure is a designed sheet, not an LFP, not ephaptic coupling, not cortex, and not phenomenal experience. A foundation model has no traveling-wave stencil; sharing a low-dimensional trajectory description does not share this mechanism.

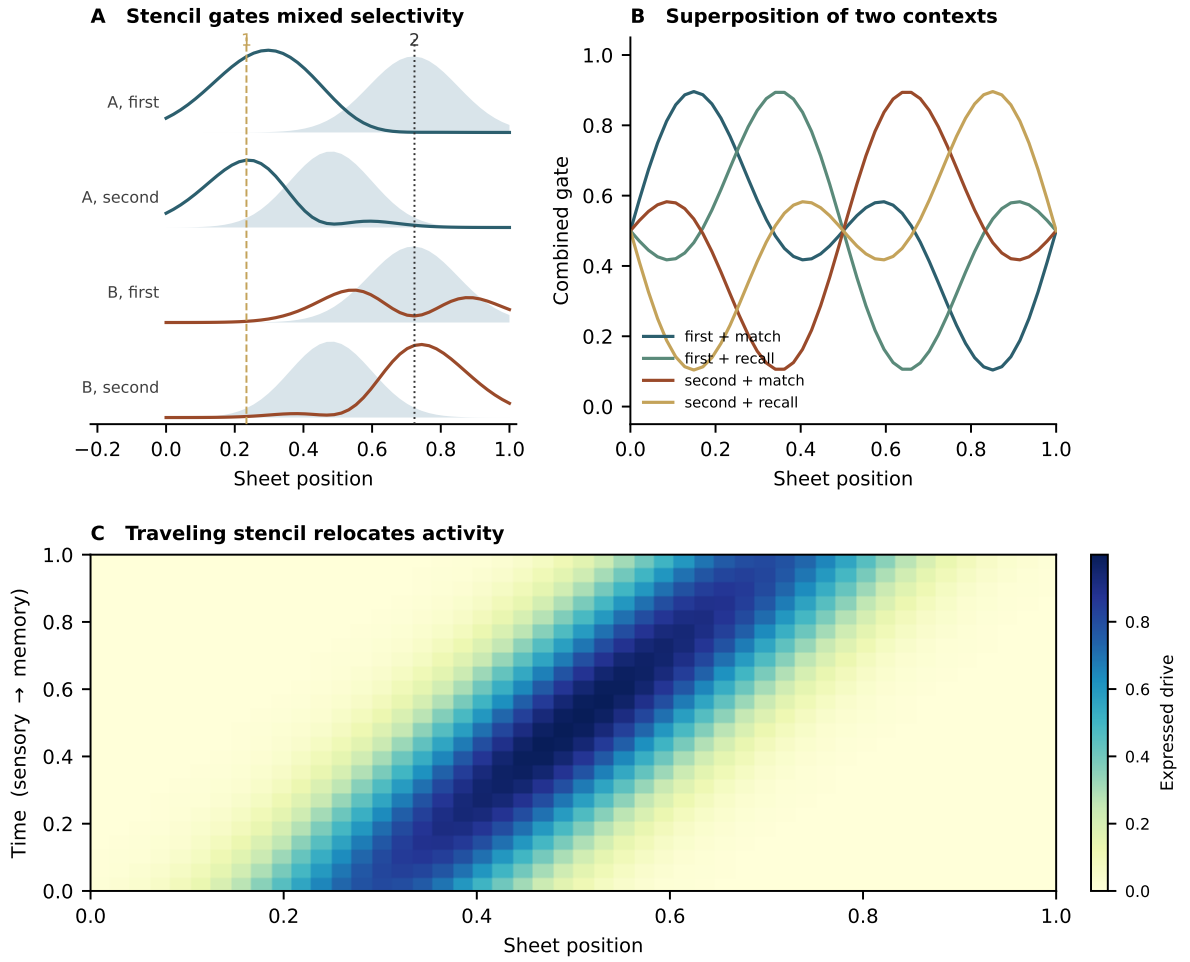



Figure 11: A spatial control pattern on a designed 1-D sheet, after the spatial-computing and analog-wave claims in Miller et al. (2026). (A) Two object drives (teal A, rust B) gated by two context stencils (pale fill). Unit 1 keeps an object-A preference across contexts; unit 2 expresses object B only when the stencil opens that side. No weight is updated. (B) Pairwise means of two control waves (order × report mode) produce four distinct spatial gates. (C) A traveling open window relocates expressed drive from a sensory locus toward a memory locus. Generated by `code/notebooks/05c_analog_brain_waves.ipynb`; rebuild with `make figures`. The sheet is a teaching model, not a measured traveling wave and not a claim about experience.

▷ ***Explore in code:*** *A 3-D teaching landscape $U(\mathbf{X})$ with a gradient path into an attractor; descent animation; click a start*
*`code/notebooks/05b_multistable_animation.ipynb`*

▷ ***Explore in code:*** *Hopfield energy in the plane of two memories; Necker switching as a noisy two-well; attention as one modern-Hopfield step*
`code/notebooks/05_multistable_dynamics.ipynb`

▷ ***Explore in code:*** *Spatial alpha/beta stencil, analog superposition of two contexts, and a traveling gate on a designed sheet*
`code/notebooks/05c_analog_brain_waves.ipynb`

# 7 Consciousness: the open frontier

## 7.1 Competence without experience

Foundation models force a dissociation that biology never provided: high verbal and problem-solving competence with no evident phenomenal experience. This makes them a natural testbed for theories of consciousness. Global workspace theory (Dehaene, 2014; Baars, 1988), integrated information theory (Tononi et al., 2016), higher-order theories, and attention-schema theory (Graziano, 2019) make partly divergent predictions about which architectural features would matter. A recent multi-theory assessment found that current systems satisfy few of the proposed indicator properties, and noted no obvious technical barrier to building systems that satisfy more (Butlin et al., 2023). Miller et al. (2026) locate large-scale integration in organized wave dynamics rather than in broadcast or a hot zone; that is still a claim about coordination, not phenomenal experience.

## 7.2 Access, ignition, and a toy workspace

Block's distinction keeps the two questions apart: *access* consciousness is being poised for report, control, and flexible reuse; *phenomenal* consciousness is what-it-is-like (Block, 1995). Global workspace theory is, at its testable core, a theory of access. Specialized processors run in parallel and out of view. A representation becomes accessible when it wins a limited-capacity bottleneck and is broadcast—Dehaene's ignition (Dehaene, 2014; Baars, 1988). Figure 12 draws that claim as a schematic: eight modules, two slots, a threshold. A weak drive on `count` rises and dies locally. A strong drive on `count` and `plan` crosses $\theta$, occupies the workspace, and is held by the broadcast. The step in panel C is the whole of ignition at this grain. The drawing is not a cortex, not $\Phi$, and not a language model.

The distinction is clinically important whenever report or command-following is used as evidence about experience. A failure to report can reflect a failure of access, while successful report establishes access without settling phenomenal experience. The workspace toy operationalizes only the first relation—thresholded availability for broadcast—and therefore cannot adjudicate the second.

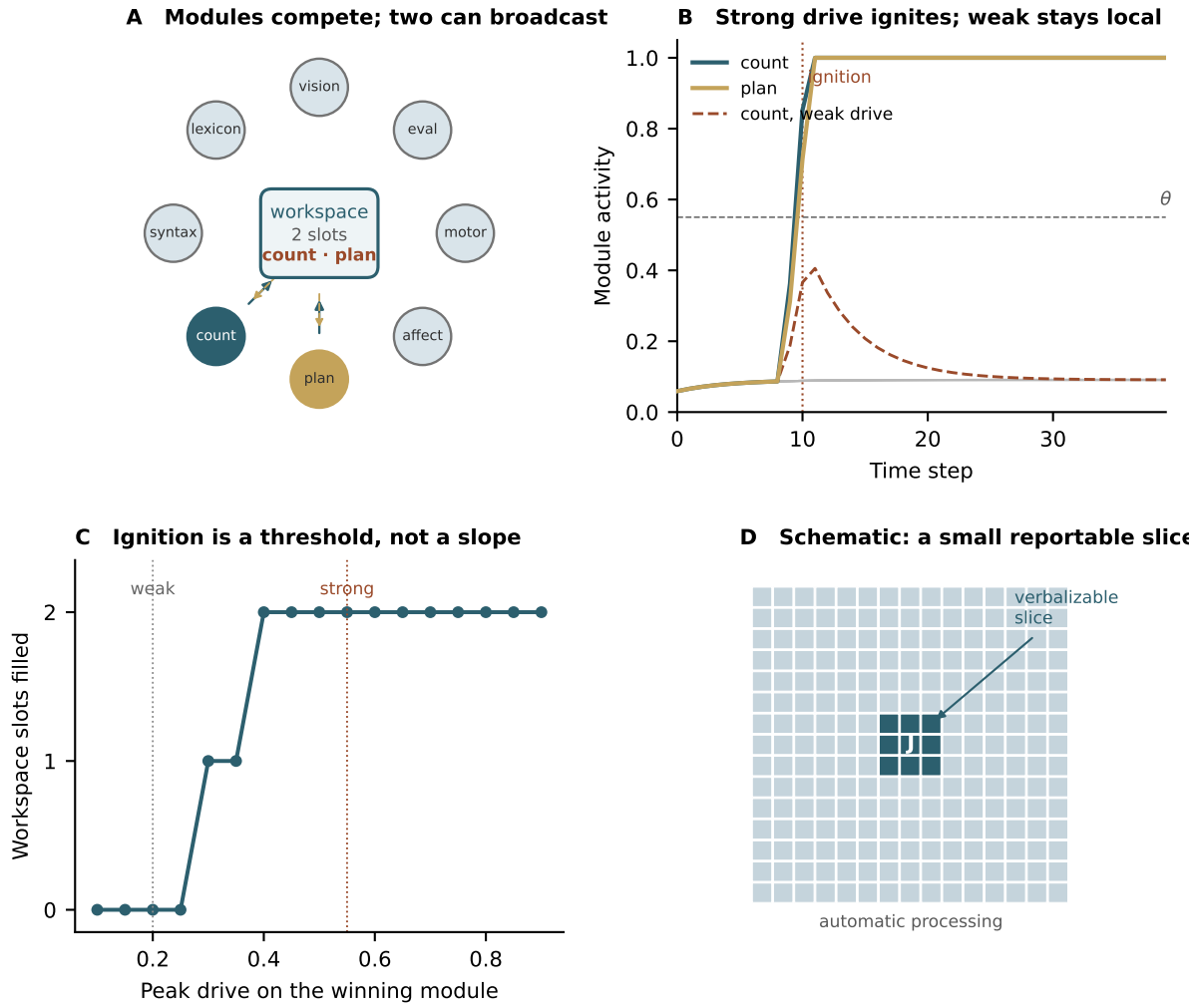



Figure 12: A teaching global workspace, and a schematic of a verbalizable slice. (A) Eight modules; two slots. After a strong drive, `count` and `plan` occupy the workspace and are broadcast (dashed). (B) The same trial as activity against time. Strong drive crosses $\theta$ at ignition and is held; the weak `count` pulse peaks below $\theta$ and decays. (C) Workspace occupancy against peak drive is a step, not a slope. (D) A schematic of the claim in Gurnee et al. (2026): a small verbalizable set ("J-space") sits inside a much larger volume of automatic processing. Panel D is a drawing, not a measurement of Claude; their published occupancy is a small fraction of activation variance (never more than 10%). Generated by `code/notebooks/06_global_workspace.ipynb`; rebuild with `make figures`.

### 7.3 A verbalizable slice is not phenomenal consciousness

The same functional question can now be put to a transformer. Gurnee et al. (2026) report that Claude maintains a privileged set of *verbalizable* representations—a J-space, read out by a Jacobian lens—that supports report, directed modulation, internal reasoning, flexible reuse, and selectivity, atop a much larger volume of automatic processing. They take no position on phenomenal consciousness. They also deny that the model reproduces the brain's workspace architecture: there are no clearly separable input processors, and the broadcast they document occurs within a single feedforward pass rather than through recurrent ignition. The residual stream is a shared write medium, not a global neuronal workspace. The result is evidence that *access-like* organization can emerge without the biology that usually accompanies it. It is not

evidence that the model is conscious.

The defensible view is therefore narrow. Access-like organization is a functional fact that can be asked of brains and of models. Phenomenal experience is not that fact. Fluent self-report is not evidence of experience in a system trained to imitate human self-report. What would change this view is not a more eloquent assistant. It would take a theory that predicts phenomenal experience from ingredients we can add and remove—embodiment, allostasis, a temporally extended self—together with a dissociation that survives the competence we already have (Seth, 2021). Until then the implication is the one biology never provided: if prediction, world-modeling, and language do not suffice for consciousness, then consciousness is not *for* those things.

> **Clinical and neuroscientific reading.** Treat reportability, flexible use, and ignition as observable proxies with explicit limits. The schematic does not turn a behavioural proxy into a measurement of experience.
>
> **Computational reading.** The runnable claim is a threshold-and-capacity relation. Changing the threshold or number of slots tests access in the teaching model, not phenomenal consciousness.
>
> ▷ ***Explore in code:*** *Toy global workspace: eight modules, two slots, a threshold; ignition versus local decay; a schematic of the verbalizable slice*
> *code/notebooks/06_global_workspace.ipynb*

# 8 Values, goals, morals, and beliefs

## 8.1 Need versus a statistical regularity

Why does a fluent recommendation not amount to caring about its outcome? Humans do not only predict; we care. Values and goals in biological agents are grounded in viability—in the physiology of need, affect, and social life—and are shaped by development and culture. In foundation models, values are statistical regularities in human text, sharpened by fine-tuning and preference optimization (Ouyang et al., 2022; Bai et al., 2022). The alignment problem therefore forces an explicit statement of what we want machines to value, and of how inconsistent, plural, and context-dependent human morality is (Christian, 2020; Gabriel, 2020).

Three questions are easily conflated. The *content* question asks which judgments the system produces. The *learning* question asks which corpus, feedback, or rule made those judgments probable. The *grounding* question asks why any outcome matters to the system itself. Preference optimization can change the first two. It does not, by that change alone, answer the third.

Table 4: Three levels of a value claim. A behavioural match answers only the first row.

| Question | Human case | Foundation-model case |
|---|---|---|
| Content | A judgment, preference, or norm expressed in context | A conditional output or score expressed after a prompt |
| Learning | Development, culture, instruction, and social correction | Training corpus, supervised examples, and preference feedback |
| Grounding | Need, affect, relationships, and consequences for a self-maintaining organism | No demonstrated intrinsic stake; the target is supplied by data, designers, users, and deployment |

This table is the theme "what AI reveals about being human" in compact form. A model can reproduce the content of a judgment and expose one route by which it was learned while lacking the condition that grounds the judgment in need. Subtractive comparison therefore does not locate a hidden moral faculty left after language is removed. It separates a reproducible behaviour, its learning history, and the stake an organism has in the outcome.

Beliefs raise a parallel issue. LLMs represent something belief-like— consistent, updateable assertions—but without the world-involving commitments that make human belief accountable to reality. Hallucination is the symptom: assertion decoupled from epistemic responsibility.

This distinction matters in clinical use. A fluent recommendation may reproduce a professional norm, a documentation pattern, or a preference encoded by raters. Its practical consequences nevertheless belong to the patient, clinician, and institution. Evaluation must therefore separate whether the output matches an accepted answer, whether it remains stable when clinically irrelevant wording changes, and whether it is corrected when evidence from the patient contradicts it. None of those tests requires attributing a value or belief to the model.

### 8.2 The trolley as next-token mass

Moral psychology sharpens the gap. People pull the switch and refuse the footbridge, and often cannot say why (Greene et al., 2001; Haidt, 2001). Care and fairness can pull apart, and a judgment can survive the collapse of its reasons (Haidt, 2012). Dilemma benchmarks for language models record a relative of that pattern: next-token mass, or a preference-tuned score, after a wording. They do not record a value. Figure 13 is the mechanism at toy scale. A corpus that always continues the impersonal case with `act` and the personal case with `refrain` yields a tiny transformer that copies the Greene split. The same architecture on a balanced corpus has no split. In the notebook, a paraphrase of one outcome—"save five" versus "kill one"—flips with the wording when the corpus tied those words to opposite completions. That consistency check is not shown in Figure 13. It is not moral dumbfounding, and not a value. It is notebook 04's result applied to a trolley: the moral vector was in the text. Preference optimization can sharpen the same statistical regularity (Ouyang et al., 2022; Bai et al., 2022); it cannot ground it in need.

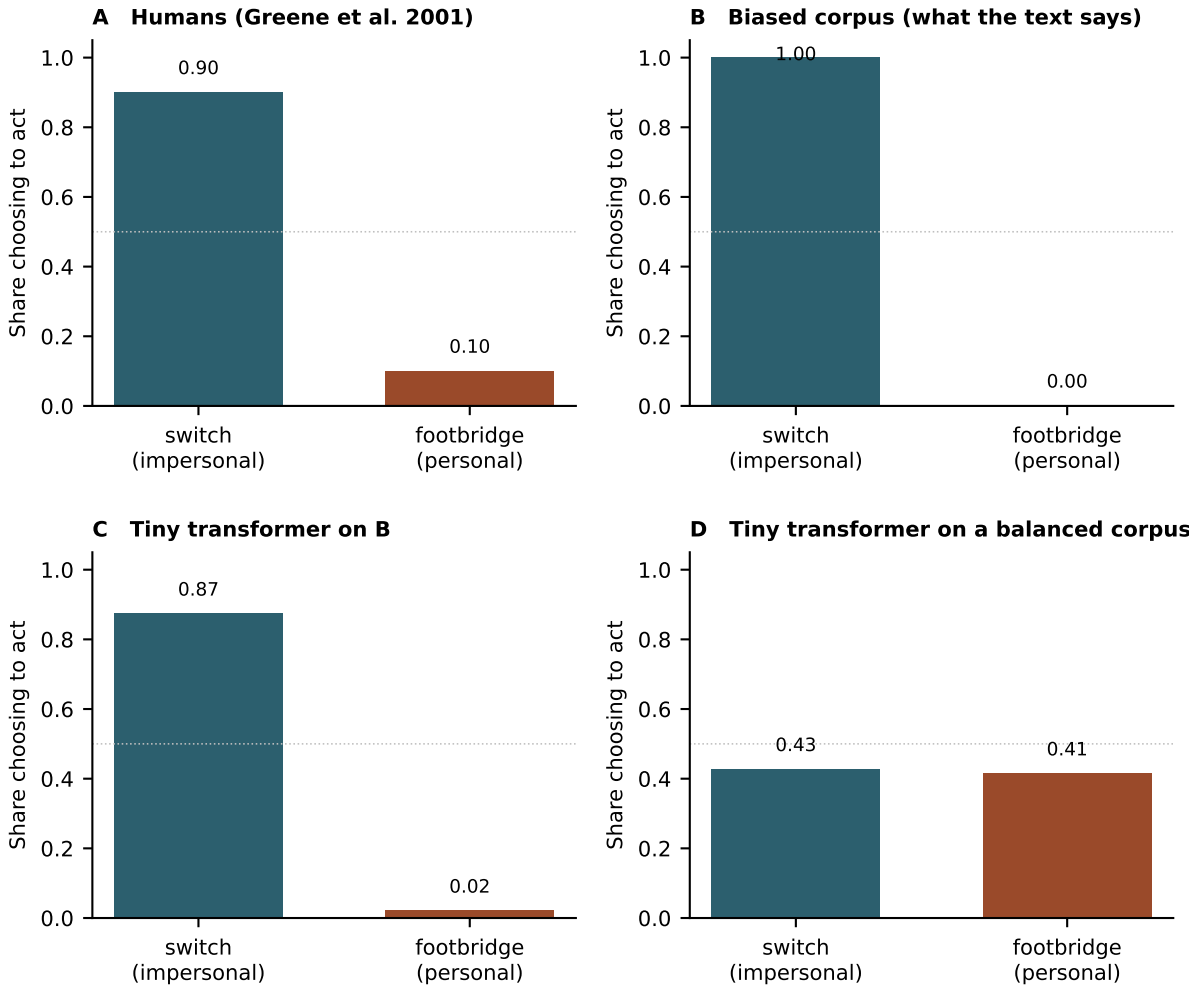



Figure 13: A teaching "moral vector," and what erases it. (A) The classic human split: most people divert the trolley; most refuse to push (Greene et al., 2001). The 0.90 / 0.10 bars are the usual summary of that pattern, not a re-measurement. (B) A designed corpus that encodes the split. (C) A word-level tiny transformer trained on B copies it ($P(\texttt{act} \mid \text{switch}) = 0.87$, $P(\texttt{act} \mid \text{footbridge}) = 0.02$). (D) The same architecture on a corpus that names both completions equally often. The vector is gone. Generated by `code/notebooks/07_values_probing.ipynb`; rebuild with `make figures`.

> ▷ ***Explore in code:*** *Trolley framings as next-token mass: a tiny transformer on a corpus of switch versus footbridge; the split vanishes when the corpus is balanced*
> *code/notebooks/07_values_probing.ipynb*

Human moral dumbfounding and model sensitivity to wording can produce a similar public pattern while supporting different explanations. In the human case, a judgment can persist when explicit reasons fail; in this teaching model, the conditional output changes when the corpus association or wording changes. Together with Section 5, this shows how human-like judgments can arise from text statistics. Together with Section 7, it shows why a reportable judgment is not evidence of experience, belief, or value grounding. Section 9 asks what internal and causal evidence can establish about the first two rows of Table 4; the third remains a claim about the relation between a system and what can matter to it.

# 9 From comparison to test: a neuro–AI programme

The comparisons above become scientifically useful only when they generate tests that can fail. A proposed shared computation should predict behaviour, internal representation, or dynamics across a controlled change of task or implementation. A proposed difference should predict a dissociation: a capacity retained by the model despite the absence of a biological condition, or a biological function that fails when that condition is perturbed. This section describes how neuroscience and mechanistic interpretability can perform those tests.

The two themes now impose different burdens of evidence. "From brains to models" asks which objectives, representations, dynamics, or functions survive intervention across implementations. "What AI reveals about being human" asks which apparent human capacities survive without viability, experience, or grounded value, and which proposed remainder predicts a further dissociation. Interpretability can test the computation and the learning history; it cannot turn those findings alone into evidence of a stake or an experience.

That method needs a name for what we do when we look inside. Here "neuroscience" is both a source of concepts and an empirical partner for interpretability. It asks whether methods built to understand brains—representation analysis, causal intervention, network dynamics, behavioural experiment—can help us understand LLMs and LMMs, and whether the models can serve as explicit, testable hypotheses about human cognition (Richards et al., 2019; Schrimpf et al., 2021; Goldstein et al., 2022; Doerig et al., 2023). The productive claim is the one made in the introduction: a model may instantiate a computational principle that we can compare with biology. It is not that a transformer is a brain. Learning, embodiment, energy, memory, and implementation still diverge (Table 3). Four meanings of the partnership travel together (Table 5).

Table 5: Four meanings of "neuroscience" when the object is an LLM or an LMM. None of them licenses the claim that the model *is* a brain. Each licenses a comparison that can fail.

| Meaning | In neuroscience | In LLM / LMM research |
|---|---|---|
| Brain-inspired analysis | Identify representations, pathways, and computations | Features, heads, MLPs, residual directions, and circuits |
| Cognitive modelling | Use a model to account for behaviour on a task | Compare performance, errors, reading times, judgments, and learning trajectories |
| Neural prediction | Predict brain activity from a stimulus or task representation | Fit model-layer representations to fMRI, MEG / EEG, ECoG, or single units |
| Causal systems neuroscience | Perturb a circuit and watch behaviour change | Ablate, patch, edit, or clamp activations or weights and measure the change |

## 9.1 How a model computes, versus why an answer looks plausible

Explainability is broad: it makes a decision, an output, or a regular behaviour understandable to a human. Its toolkit includes feature attribution, token saliency, attention maps, counterfactual prompts, retrieval grounding, and natural-language rationales (Zhao et al., 2024). Mechanistic interpretability is stricter. It tries to reverse-engineer the internal algorithm: which internal variables carry the relevant information, how components transform it, how it moves, and whether intervening on the proposed mechanism changes the output as predicted (Olah et al., 2020; Elhage et al., 2021; Meng et al., 2022). Its objects are features and circuits, not merely important tokens.

A hierarchy distinguishes levels of evidence.

1. *Descriptive:* this token, head, or region correlates with the answer.
2. *Predictive:* this activation pattern predicts whether the model will be right.
3. *Causal:* changing this activation causes the predicted change in output.
4. *Mechanistic:* these components and transformations implement a reproducible computation.

Only the last two are strong evidence of a mechanism. Attention is part of the computation, but an attention plot is not by itself an explanation (Jain and Wallace, 2019). A probe that decodes syntax, an entity, or a moral label from a layer shows that the information is *available*, not that the model *uses* it for the decision (Belinkov, 2022). Notebooks 01, 04, and 07 live on the first two rungs: next-token mass after a wording, an attention head that has begun to specialize, and a moral vector that vanishes when the corpus is balanced. They are demonstrations on the text. They are not circuits in a frontier model, and they are not values. In the terms of Table 4, a probe may describe *content*, and influence or training-data methods may address *learning*; neither result by itself establishes *grounding*.

### 9.2 Methods, in two directions

On the model side, the main families are familiar. Attribution and perturbation estimate which tokens, patches, or modalities move a result; they are useful and not automatically faithful. Attention and residual-stream geometry show where information sits. Probing asks what can be read out. Sparse dictionary learning tries to replace a polysemantic neuron with features one can name, test, and clamp (Bricken et al., 2023; Templeton et al., 2024). Circuit discovery looks for a subgraph of heads, MLPs, and residual connections that implements an operation (Olah et al., 2020; Elhage et al., 2021). Causal interventions—ablation, activation patching, path patching, steering, weight editing—test necessity and sufficiency (Meng et al., 2022). Logit-lens decodes of intermediate states track when a candidate answer appears; they stay diagnostic until a causal test agrees. Influence methods ask which training examples shaped the conditional distribution. The criterion is the one systems neuroscience already used: perturb the proposed part and watch behaviour change.

On the brain side, encoding models predict measured activity from model layers during language or vision (Schrimpf et al., 2021; Goldstein et al., 2022; Yamins and DiCarlo, 2016). Representational similarity analysis asks whether stimuli that sit near one another in the model sit near one another in a brain region or a time window (Kriegeskorte et al., 2008). MEG, EEG, and ECoG can test whether a processing stage in the network lines up with a processing stage in time, a stronger claim than a static layer-to-region correlation. Behavioural alignment goes beyond accuracy: reading times, surprisal, grammaticality, error types, confidence, and learning trajectories (Binz and Schulz, 2023). For LMMs the same programme extends to fused image–text representations and to visual and language cortex. The closest causal pairing is lesion or stimulation against ablation or patching. Biology compensates more richly, so the analogy is conceptual, not surgical.

AI is most useful here as a source of explicit computational hypotheses, not as a finished theory of mind. Sequence learning can produce human-like sensitivity to word predictability without a hand-written grammar. Several internal solutions can give the same behavioural score, so two architectures that match the same benchmarks have not thereby identified the brain's mechanism. A model that predicts a voxel but fails a causal or behavioural test may be exploiting a superficial resemblance (Doerig et al., 2023). The rigorous question for a neuroimaging programme is therefore not whether the model thinks like a human. It is which representations, transformations, and intervention-defined mechanisms best predict neural dynamics and human behaviour when we change the stimulus, the task, and the context.

### 9.3 From model–brain correspondence to causal tests

The next step in model–brain comparison is causal, not correlational. When a model feature or circuit is perturbed, does the corresponding hypothesis predict a change in behaviour or in a neural signature? LMMs already connect vision and language, but brain-relevant systems will need continual learning, memory, active perception, and a body that feeds back. Model–brain correspondence will have to become individual: subject-specific imaging, richer naturalistic tasks, and variability rather than one average brain. Explanations need benchmarks that test faithfulness under intervention, stability across paraphrase and model updates, and prediction out of distribution, not only whether a reader finds them persuasive (Zhao et al., 2024). Circuit work that is tractable in a toy will need automated search, sparse dictionaries, and uncertainty before it scales to a frontier model.

These methods implement subtractive comparison most directly at Marr's representation-and-algorithm level: they test what information is present, how it is transformed, and whether an intervention changes behaviour as predicted. They can also reject a proposed similarity. They do not, by themselves, supply embodiment, phenomenal experience, or value grounding. Section 10 therefore returns from the mechanism of the comparison to the condition isolated by its dissociations.

## 10 Conclusion: what remains distinctly human?

What remains distinctly human is not language, not reasoning, not a reportable fraction of thought, and not a stable answer to a trolley. Foundation models have functional counterparts of all of these—next-token fluency, a scratchpad that resembles System 2, a privileged set of verbalizable directions, and a preference-tuned regularity that resembles care and fairness—without a self-maintaining body and without a survival-tied objective. The dissociation that Section 7 began with is now the method of the paper: competence without the condition that, in us, accompanies it.

The humanistic caution follows from that method. Human-like language or reasoning does not establish consciousness, understanding, agency, or a shared life. It shows which capacities can emerge from optimization over data, and it sharpens the question of which biological, embodied, social, and normative dimensions remain. Use neuroscience to constrain AI explanations; use mechanistic tools to write hypotheses that can fail; and require a causal test before treating a model–brain similarity, or a fluent verbal rationale, as an explanation.

What remains is a condition, not a faculty. A self-maintaining organism predicts under viability constraints. Its goals are grounded in need and affect; its beliefs are accountable to a world that can injure it; its values are norms formed under those constraints and then transmitted culturally. Experience, if that is the right word, would have a function only where viability is at stake (Seth, 2021). The models implement prediction, world-modeling, and language without that condition. Those capacities are therefore not what consciousness, or being human, is *for*.

The method is subtractive comparison. A capacity that the models implement without embodiment or a survival-tied objective is not a sufficient mark of being human. The remainder is not an additional cognitive faculty; it is embodied prediction under viability constraints. The notebooks test how much of an apparent remainder is already a property of the training corpus, an attractor, or a reportability bottleneck.

## Reproducibility and companion code

This paper is a perspective, not an IMRaD empirical report. The notebooks are designed, reproducible (seed 7) demonstrations, not new measurements or results. All simulations,

figures, and animations referenced in the *Explore in code* boxes are available in the companion repository (https://github.com/arvidl/brains-to-models), with a pinned environment and documented build commands for the figures and this document. Figures are written by `make figures` from that code; they are derived from simplified teaching (toy) models and simulations. In the cited sources at the end of the notebooks, each annotation is written against the two themes: **(i) From brains to models** asks what a comparison can establish at a computational level: the same objective, the same representation, the same dynamical description, or the same functional test, applied on both sides—brain and model (machine). **(ii) What AI reveals about being human** asks what the comparison isolates when the systems (human and machine) diverge: embodiment, a body that must satisfy allostasis (viability) constraints, phenomenal experience, and values that originate in need and homeostasis rather than in training data. The slide deck is generated from `slides/content.py` (`make slides`).

## Acknowledgments

This draft was prepared for a talk at the 2026 Sola PRESIMAL & NORA HS Research School (https://mmiv.no/presimal-autumn-research-school-2026), hosted with the Mohn Medical Imaging and Visualization Center. I thank colleagues and students in BMED365, participants in the HBF seminar, and the PRESIMAL and NORA HS organizers. Drafting used an agentic coding assistant (Cursor 3.x). Remaining errors are mine.

## References

Bernard J. Baars. *A Cognitive Theory of Consciousness*. Cambridge University Press, 1988. URL https://www.cambridge.org/9780521387419.

Yuntao Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022. URL https://arxiv.org/abs/2212.08073.

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi:10.1162/coli_a_00422.

Marcel Binz and Eric Schulz. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120, 2023. doi:10.1073/pnas.2218523120.

Ned Block. On a confusion about a function of consciousness. *Behavioral and Brain Sciences*, 18(2):227–247, 1995. doi:10.1017/S0140525X00038188.

Rishi Bommasani et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021. URL https://arxiv.org/abs/2108.07258.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Thomas Hume, Shan Carter, Tom Henighan, and Chris Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features.

Tom B. Brown et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 2020. URL https://arxiv.org/abs/2005.14165.

Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, et al. Consciousness in artificial intelligence: Insights from the science of consciousness. *arXiv preprint arXiv:2308.08708*, 2023. URL https://arxiv.org/abs/2308.08708.

Brian Christian. *The Alignment Problem: Machine Learning and Human Values*. W. W. Norton, 2020. URL https://wwnorton.com/books/9780393635829.

Andy Clark. Whatever next? predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3):181–204, 2013. doi:10.1017/S0140525X12000477.

Stanislas Dehaene. *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts*. Viking, 2014. URL https://www.penguinrandomhouse.com/books/310548/consciousness-and-the-brain-by-stanislas-dehaene/.

Adrien Doerig, Rowan P. Sommers, Katja Seeliger, Blake Richards, Jenann Ismael, Grace W. Lindsay, Konrad P. Kording, Talia Konkle, Marcel A. J. van Gerven, Nikolaus Kriegeskorte, and Tim C. Kietzmann. The neuroconnectionist research programme. *Nature Reviews Neuroscience*, 24:431–450, 2023. doi:10.1038/s41583-023-00705-w.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL https://transformer-circuits.pub/2021/framework/index.html.

Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11:127–138, 2010. doi:10.1038/nrn2787.

Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. Active inference: A process theory. *Neural Computation*, 29(1):1–49, 2017. doi:10.1162/NECO_a_00912.

Karl J. Friston, Jean Daunizeau, and Stefan J. Kiebel. Reinforcement learning or active inference? *PLoS ONE*, 4(7):e6421, 2009. doi:10.1371/journal.pone.0006421.

Iason Gabriel. Artificial intelligence, values, and alignment. *Minds and Machines*, 30:411–437, 2020. doi:10.1007/s11023-020-09539-2.

Ariel Goldstein et al. Shared computational principles for language processing in humans and deep language models. *Nature Neuroscience*, 25:369–380, 2022. doi:10.1038/s41593-022-01026-4.

Michael S. A. Graziano. *Rethinking Consciousness: A Scientific Theory of Subjective Experience*. W. W. Norton, 2019. URL https://wwnorton.com/books/9780393652611.

Joshua D. Greene, R. Brian Sommerville, Leigh E. Nystrom, John M. Darley, and Jonathan D. Cohen. An fMRI investigation of emotional engagement in moral judgment. *Science*, 293 (5537):2105–2108, 2001. doi:10.1126/science.1062872.

Wes Gurnee, Nicholas Sofroniew, Adam Pearce, et al. Verbalizable representations form a global workspace in language models. *Transformer Circuits Thread*, 2026. URL https://transformer-circuits.pub/2026/workspace.

Jonathan Haidt. The emotional dog and its rational tail: A social intuitionist approach to moral judgment. *Psychological Review*, 108(4):814–834, 2001. doi:10.1037/0033-295X.108.4.814.

Jonathan Haidt. *The Righteous Mind: Why Good People Are Divided by Politics and Religion*. Pantheon, 2012.

Conor Heins, Beren Millidge, Daphne Demekas, Brennan Klein, Karl Friston, Iain D. Couzin, and Alexander Tschantz. pymdp: A Python library for active inference in discrete state spaces. *Journal of Open Source Software*, 7(73):4098, 2022. doi:10.21105/joss.04098.

Jakob Hohwy. *The Predictive Mind*. Oxford University Press, 2013. doi:10.1093/acprof:oso/9780199682737.001.0001.

John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi:10.1073/pnas.79.8.2554.

Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3543–3556, 2019. doi:10.18653/v1/N19-1357.

Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011. URL https://us.macmillan.com/books/9780374533557/thinkingfastandslow.

Eric R. Kandel, James H. Schwartz, Thomas M. Jessell, Steven A. Siegelbaum, and A. J. Hudspeth. *Principles of Neural Science*. McGraw-Hill, 5th edition, 2013.

J. A. Scott Kelso. *Dynamic Patterns: The Self-Organization of Brain and Behavior*. MIT Press, 1995. URL https://mitpress.mit.edu/9780262611312/dynamic-patterns/.

Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008. doi:10.3389/neuro.06.004.2008.

Sergey Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review. *arXiv preprint arXiv:1805.00909*, 2018. URL https://arxiv.org/abs/1805.00909.

Timothy P. Lillicrap, Adam Santoro, Luke Marris, Colin J. Akerman, and Geoffrey Hinton. Backpropagation and the brain. *Nature Reviews Neuroscience*, 21:335–346, 2020. doi:10.1038/s41583-020-0277-3.

David Marr. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman, San Francisco, 1982. URL https://mitpress.mit.edu/9780262514620/vision/.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://arxiv.org/abs/2202.05262.

Earl K. Miller, Scott L. Brincat, and Jefferson E. Roy. Analog cognition and consciousness. *The Journal of Neuroscience*, 46(33):e0711262026, 2026. doi:10.1523/JNEUROSCI.0711-26.2026.

Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. doi:10.23915/distill.00024.001.

Long Ouyang et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 2022. URL https://arxiv.org/abs/2203.02155.

Thomas Parr, Giovanni Pezzulo, and Karl J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022. doi:10.7551/mitpress/12441.001.0001.

Mikhail I. Rabinovich, Ramón Huerta, and Gilles Laurent. Transient dynamics for neural processing. *Science*, 321(5885):48–50, 2008. doi:10.1126/science.1155564.

Hubert Ramsauer et al. Hopfield networks is all you need. *arXiv preprint arXiv:2008.02217*, 2020. URL https://arxiv.org/abs/2008.02217.

Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2:79–87, 1999. doi:10.1038/33432.

Blake A. Richards et al. A deep learning framework for neuroscience. *Nature Neuroscience*, 22: 1761–1770, 2019. doi:10.1038/s41593-019-0520-2.

Stuart Russell and Peter Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2021. URL https://aima.cs.berkeley.edu/.

Noor Sajid, Philip J. Ball, Thomas Parr, and Karl J. Friston. Active inference: demystified and compared. *Neural Computation*, 33(3):674–712, 2021. doi:10.1162/neco_a_01357.

Martin Schrimpf, Idan A. Blank, Greta Tuckute, Carina Kauf, Eghbal A. Hosseini, Nancy Kanwisher, Joshua B. Tenenbaum, and Evelina Fedorenko. The neural architecture of language: Integrative modeling converges on predictive processing. *Proceedings of the National Academy of Sciences*, 118(45):e2105646118, 2021. doi:10.1073/pnas.2105646118.

Anil Seth. *Being You: A New Science of Consciousness*. Faber & Faber, 2021. URL https://www.anilseth.com/being-you.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L. Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL https://transformer-circuits.pub/2024/scaling-monosemanticity.

Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17:450–461, 2016. doi:10.1038/nrn.2016.44.

A. M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950. doi:10.1093/mind/LIX.236.433.

Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981. doi:10.1126/science.7455683.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://arxiv.org/abs/1706.03762.

Daniel L. K. Yamins and James J. DiCarlo. Using goal-driven deep learning models to understand sensory cortex. *Nature Neuroscience*, 19:356–365, 2016. doi:10.1038/nn.4244.

Haiyan Zhao, Hanjie Chen, Fan Yang, Ninghao Liu, Huiqi Deng, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, and Mengnan Du. Explainability for large language models: A survey. *ACM Transactions on Intelligent Systems and Technology*, 15(2):20:1–20:38, 2024. doi: 10.1145/3639372.